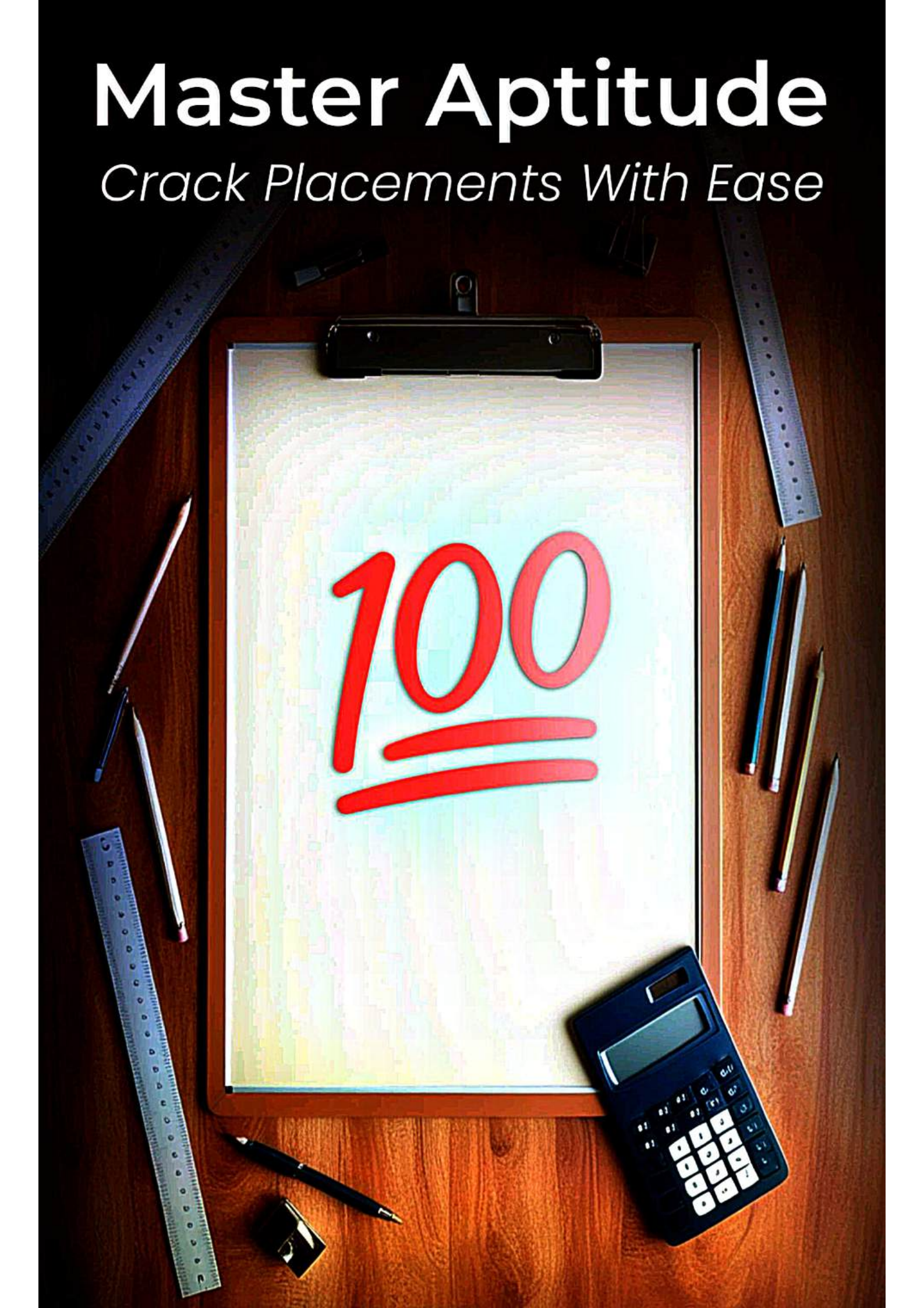


Master Aptitude

Crack Placements With Ease



100

MASTER APTITUDE

Crack Placements With Ease

Introduction

Welcome to Master Aptitude: Crack Placements With Ease, your comprehensive guide to conquering quantitative aptitude for placement exams and competitive tests. This book is designed to equip you with the essential skills and techniques to solve quantitative aptitude questions efficiently and effectively, whether you're a beginner looking to build your foundation or an experienced learner aiming to refine your skills.

Quantitative aptitude tests are a crucial part of almost every competitive examination and campus placement. These tests assess your problem-solving ability, numerical skills, and analytical thinking—all essential qualities for succeeding in any professional field. Yet, many students find quantitative aptitude challenging, often due to a lack of practice or understanding of fundamental concepts. In this book, we aim to break down complex problems into manageable steps, using clear explanations and examples that help you understand the core principles behind each topic.

From basic arithmetic and algebra to advanced problem-solving strategies, this guide covers it all. You'll find tips, shortcuts, and techniques specifically tailored to help you save time and avoid common pitfalls during exams.

Here's what you can expect in Master Aptitude:

- **Foundational Concepts:** We'll start with the basics, ensuring that you have a strong understanding of key concepts in arithmetic, algebra, geometry, and data interpretation.

- **Practice Questions:** A wide variety of practice problems with detailed solutions will help reinforce each concept and build your confidence.
- **Test-Taking Strategies:** Learn practical techniques for tackling difficult questions, managing time, and staying calm under pressure.
- **Real-World Applications:** Understand how quantitative aptitude skills are useful not just in exams but also in real-life situations and career-related problem-solving.

Whether you're preparing for campus placements, competitive exams, or simply looking to sharpen your quantitative skills, this book is here to guide you every step of the way. Remember, quantitative aptitude is not just about knowing formulas; it's about developing a logical and analytical approach to problem-solving.

We hope this book empowers you to approach aptitude tests with confidence and ease, giving you the tools to achieve the scores you're aiming for. Now, let's dive in and start your journey to mastering quantitative aptitude!

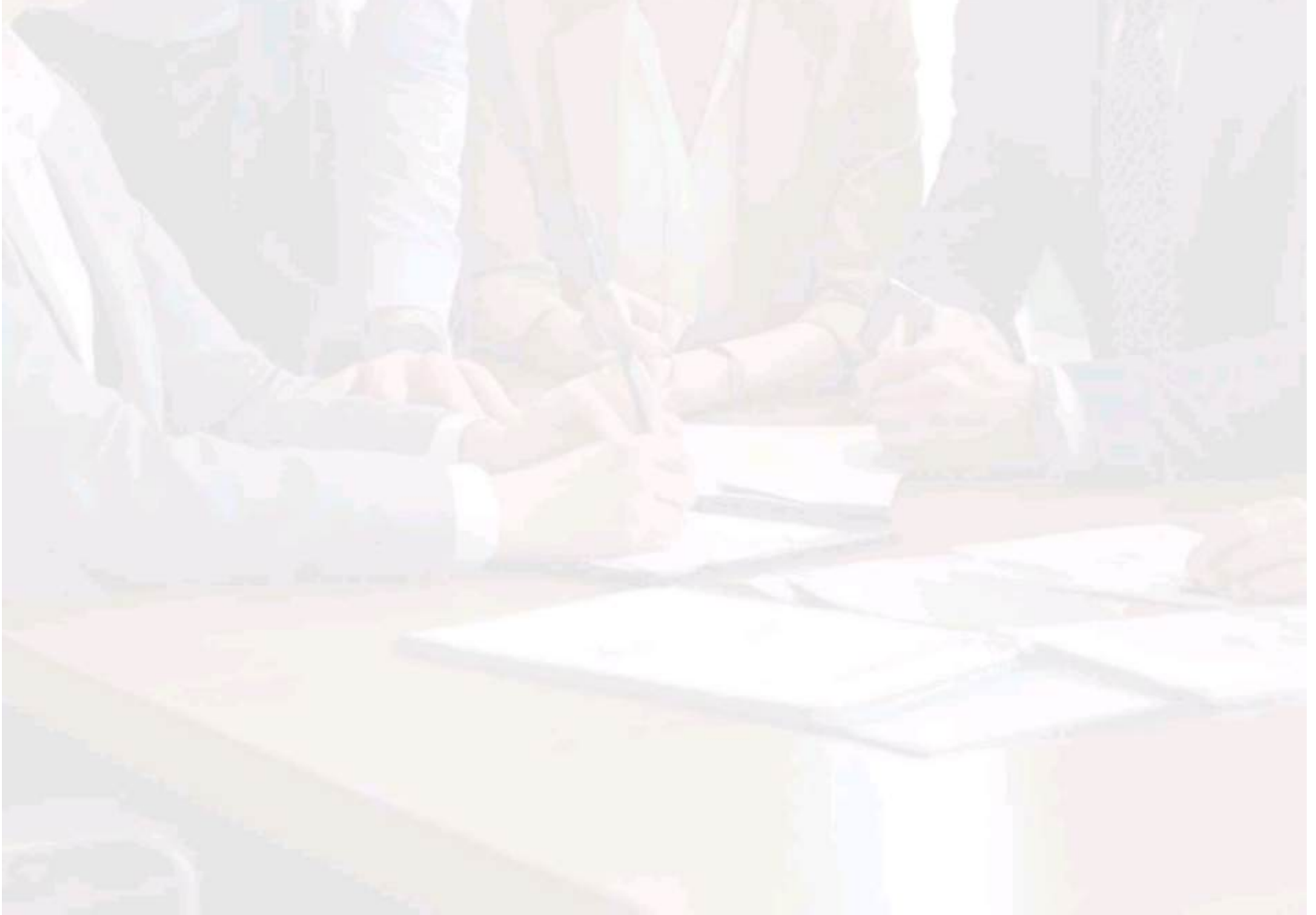


Table Of Contents

Quantitative Aptitude

Number System	1
Averages	15
Percentages	20
Ratio Proportion	25
Chain Rule	29
Profit, Loss and Discounts	34
Mixture and Allegations	39
Simple and Compound Interest	45
Time and Distance	50
Time and Work	54
Permutation and Combination	58
Probability	63
Mensuration	68
Data Interpretation	79
Algebra	88

Placement Papers

Placement Paper 1	92
Placement Paper 2	99

Table Of Contents

Placement Papers

Placement Paper 3	106
Placement Paper 4	112
Placement Paper 5	118

Career Guidance

Tips for Applying Jobs	123
HR Interviews	125
Resume Preparation	127
Group Discussion	129

Number System

In Hindu–Arabic system we use ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 called digits to represent any number. This is the decimal system where we use the digits 0 to 9. Here 0 is called insignificant digit where as 1, , 9 are called significant digits.

• Classification of Numbers:

Natural Numbers: The numbers 1, 2, 3, 4, 5, 6, which we use in counting are known as natural numbers. The set of all natural numbers can be represented by $N = \{1, 2, 3, 4, 5,\}$

Whole Numbers: If we include 0 among the natural numbers then the numbers 0, 1, 2, 3, 4, 5, are called whole numbers. Hence, every natural number is a whole number. The set of whole numbers is represented by W.

Integers: All counting numbers and their negatives including zero are known as integers. The set of integers can be represented by Z or I. $Z = \{. -4, -3, -2, -1, 0, 1, 2, 3, 4,\}$. Every natural number is an integer but every integer is not natural number.

Positive Integers: The set $I^+ = \{1, 2, 3, 4,\}$ is the set of all positive integers. Positive integers and Natural numbers are synonyms.

Negative Integers: The set $I^- = \{. . . , -3, -2, -1\}$ is the set of all negative integers. 0 (zero) is neither positive nor negative.

Non Negative Integers: The set $\{0, 1, 2, 3, . . .\}$ is the set of all non negative integers.

Rational Numbers: The numbers of the form $\frac{p}{q}$ where p and q

are integers, p is not divisible by q and $q \neq 0$, are known as rational numbers. **(or)** Any number that can be written in fraction form is a rational number. This includes integers, terminating decimals, and repeating decimals as well as fractions.

$\frac{3}{7}, \frac{5}{2}, -\frac{5}{9}, \frac{1}{2}, -\frac{3}{5}$, etc. The set of rational numbers is denoted by Q.

Irrational Numbers: Any real number that cannot be written in fraction form is an irrational number. Numbers which are both non-terminating as well as non repeating decimals are called

irrational numbers. e.g.: Absolute value of $\frac{10}{3}, \frac{22}{7}, \sqrt{2}, \sqrt{3}, \sqrt{10}...$

Note: A terminating decimal will have a finite number of digits after the decimal point.

e.g.: $\frac{3}{4} = 0.75, \frac{5}{4} = 1.25, \frac{25}{16} = 1.5625$.

Repeating Decimals: A decimal number that has digits that repeat forever. e.g.: $1/3 = 0.333...$ (here, 3 repeats forever).

Non-Repeating Decimal: A decimal that neither terminates nor repeats. e.g.: $\sqrt{2} = 1.4142135623...$

Real Numbers: The rational and irrational numbers together are called real numbers. e.g.: $13/21, 2/5, -3/7, 4/2$ etc are real numbers. The set of real numbers is denoted by R.

Even Numbers: Any integer that can be divided exactly by 2. e.g.: 2, 6, 0, -8, -10, are even numbers.

Odd Numbers: An integer that cannot be divided exactly by 2 is an Odd number. e.g.: 1, 3, -5, -7, are odd numbers.

Prime Numbers: A Prime Number can be divided evenly only by 1, or itself. And it must be a whole number greater than 1. e.g.: Numbers 2, 3, 5, 7, 11, 13, 17, are prime. All primes which are greater than 3 are of the form $(6n+1)$ or $(6n-1)$.

Note: • 1 is not a prime number.

• 2 is the least and only even prime number.

• 3 is the least odd prime number.

• Prime numbers up to 100 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97. There are 25 prime numbers up to 100.

Composite Number: Natural numbers greater than 1 which are not prime, are known as composite numbers. The number 1 is neither prime nor composite.

Co-Prime Numbers: Two numbers which have only 1 as the common factor are called co-primes (**or**) relatively prime to each other. e.g.: 3 and 5 are co primes.

Note: Natural Numbers = 1 + Prime + Composite Numbers.

Whole Numbers = 0 (Zero) + Natural Numbers.

Integers = Negative Integers + 0 + Positive Integers.

Real Numbers = Rational + Irrational Numbers.

Any two digit number is represented as $(10a + b)$, where a is the first digit, and b is the second. e.g.: 63 is represented as $10(6)+3$;

- The sum of its digits = $(a+b)$.
- The difference of the digits = $(a-b)$. Similarly, any three digit number is represented as $(100a + 10b + c)$ etc.

• Test of Divisibility:

Divisibility by 2: A number is divisible by 2 if the unit's digit is either zero or divisible by 2. e.g.: Units digit of 76 is 6 which is divisible by 2 hence 76 is divisible by 2. Units digit of 330 is 0 so it is divisible by 2.

Divisibility by 3: A number is divisible by 3 if sum of all digits in it is divisible by 3. e.g.: The number 273 is divisible by 3 since $2 + 7 + 3 = 12$ which is divisible by 3.

Divisibility by 4: A number is divisible by 4, if the number formed by the last two digits in it is divisible by 4, or last two digits are zeros. e.g.: The number 5004 is divisible by 4 since last two digits 04 is divisible by 4.

Divisibility by 5: A number is divisible by 5 if the units digit in the number is either 0 or 5. e.g.: 375 is divisible by 5 as 5 is in the units place.

Divisibility by 6: A number is divisible by 6 if it is even and sum of all digits is divisible by 3. e.g.: The number 6492 is divisible by 6 as it is even and sum of its digits $6 + 4 + 9 + 2 = 21$ is divisible by 3.

Divisibility by 7: Step-1: Remove unit's digit. And double it.

Step-2: Subtract it from the rest of the number.

Step-3: Check whether the resulted number is divisible by 7 or not.

Step-4: Repeat the above steps until the resulted number is either 0 (zero) or divisible by 7. e.g.: Consider the number 10717.

Step-1: Removing the unit's digit i.e. 7. Double of 7 = 14.

Step-2: $1071 - 14 = 1057$.

Step-3: Now remove 7 from 1057 and double it i.e. 14.

Step-4: $105 - 14 = 91$.

Step-5: Now remove 1 and double it i.e. 2. Step-6: $9 - 2 = 7$ The

final result 7 is divisible by 7. So the given number i.e. 10717 is also divisible by 7.

Divisibility by 8: A number is divisible by 8, if the number formed by last 3 digits is divisible by 8. e.g.: The number 6573392 is divisible by 8 as the last 3 digits '392' is divisible by 8.

Divisibility by 9: A number is divisible by 9 if the sum of its digit is divisible by 9. e.g.: The number 15606 is divisible by 9 as the sum of the digits $1 + 5 + 6 + 0 + 6 = 18$ is divisible by 9.

Divisibility by 10: Last digit should be zero. e.g.: Last digit of 4470 is zero. So, it is divisible by 10.

Divisibility by 11: A number is divisible by 11 if the difference of the sum of the digits at odd places and sum of the digits at the even places is either zero or divisible by 11. **(or)** Subtract the first digit from a number made by the other digits. If that number is divisible by 11 then the original number is also divisible by 11. e.g.: In the number 9823, the sum of the digits at odd places is $9+2=11$ and the sum of the digits at even places is $8+3=11$. Difference between them is $11 - 11 = 0$. Hence, the given number is divisible by 11. e.g.: 14641 $1464 - 1$ is 1463; $146 - 3$ is 143; $14 - 3 = 11$, which is divisible by 11, so 14641 is also divisible by 11.

Divisibility by 12: A number is divisible by 12 if it is divisible by 3 and 4. e.g.: The number 1644 is divisible by 12 as it is divisible by 3 and 4. Here 3 and 4 because they are co-prime to each other.

Divisibility by 13: Repeatedly add 4 times the last digit to the rest until you get a number divisible by 13 .

e.g.: $7462 \Rightarrow 746 + (2 \times 4) = 754 \Rightarrow 75 + (4 \times 4) = 91$ 91 is divisible by 13. So, 7462 is also divisible by 13.

Divisibility by 14: The number is divisible by 7 and 2.

Divisibility by 15: The number is divisible by 3 and 5.

Divisibility by 16: With a 3 digit number: Multiply hundreds digit by 4, then add the last two digits. e.g.: $352 \Rightarrow (3 \times 4) + 52 = 12 + 52 = 64$ 64 is divisible by 16. So, 352 is also divisible by 16. With a more than 3 digit number: The last four digits form a number is divisible by 16. e.g.: 38512 Here is 8512 is divisible by 16. So, 38512 is $\Rightarrow$ also divisible by 16.

Divisibility by 17: Subtract 5 times the last digit from the rest. e.g.: $3961 \Rightarrow 396 - (1 \times 5) = 391 \Rightarrow 39 - (1 \times 5) = 34$, 34 is divisible by 17. So, 3961 is also divisible by 17.

Divisibility by 18: An even number satisfying the divisibility test by 9 is also divisible by 18. e.g.: The number 80388 is divisible by 18 as it satisfies the divisibility test of 9.

Divisibility by 19: Add twice the last digit to the rest. e.g.: $10944 \Rightarrow 1094 + (4 \times 2) = 1102 \Rightarrow 110 + (2 \times 2) = 114 \Rightarrow 11 + (4 \times 2) = 19$

Divisibility by 20: Last digit is zero & tens digit is even. e.g.: 980; Last digit is zero. And tens digit is even.

Divisibility by 25: A number is divisible by 25 if the number formed by the last two digits is divisible by 25 or the last two digits are zero. e.g.: The number 7975 is divisible by 25 as the last two digits are divisible by 25.

• Common Factors:

A common factor of two or more numbers is a number which divides each of them exactly. e.g.: 3 is a common factor of 6 and 15.

• **Highest Common Factor (HCF):** Highest common factor of two or more numbers is the greatest number that divides each of them exactly. e.g.: 3, 4, 6, 12 are the factors of 12 and 36. Among

them the greatest is 12. Hence the HCF of 12, 36 is 12. HCF is also called as Greatest common divisor (GCD) or Greatest Common measure (GCM).

Method of Finding HCF: Method of division

• **HCF of Two Numbers:** Step 1: Greater no. is divided by the smaller number.

Step 2: Divisor of step - 1 is divided by its remainder.

Step 3: Divisor of step - 2 is divided by its remainder.

This could be continued until the remainder is 0. Then HCF = Divisor of the last step. e.g.: Find the HCF of 96 and 348.

Explanation: Here the divisor of the last step is 12. So, HCF of 96 and 348 is 12.

$$\begin{array}{r}
 96)248(3 \\
 \underline{288} \\
 60)96(1 \\
 \underline{60} \\
 36)60(1 \\
 \underline{36} \\
 24)36(1 \\
 \underline{24} \\
 \rightarrow 12)24(2 \\
 \underline{24} \\
 \underline{00}
 \end{array}$$

• **HCF of More than Two Numbers:**

Step 1: Take any two numbers and find their HCF.

Step 2: Now find the HCF of third number and HCF obtained for the previous two numbers.

Step 3: Now find the HCF of fourth number and HCF obtained in the previous step. Continue the same process till the last number. The final HCF is concluded to be the HCF of all the given numbers. e.g.: Find the HCF of 120, 246, 100.

$$\begin{array}{r}
 120)246(2 \\
 \underline{240} \\
 6)120(20 \\
 \underline{120} \\
 0
 \end{array}$$

6 is HCF of 120, 246. Now take 3rd number (i.e. 100) and HCF obtained in the above step (i.e. 6) and find HCF.

$$\begin{array}{r}
 6)100(16 \\
 \underline{96} \\
 4)6(1 \\
 \underline{4} \\
 \rightarrow 2)4(2 \\
 \underline{4} \\
 0
 \end{array}$$

▪ **HCF of Decimals:** e.g.: Find HCF of 3.2, 4.12, 1.3, 7.

Explanation: First eliminate the influence of decimals by multiplying it either by 10 or 100 or 1000 etc. Here multiply the numbers with 100 to make all the numbers decimal free. i.e. 320, 412, 130, 700.

Now, find the HCF of above numbers. We get it as 2. Did you remember we multiplied all the numbers by 100 to eliminate the influence of decimals. Hence, now we divide the answer by 100 to

get HCF of the original numbers. The HCF is $\frac{2}{100} = 0.02$.

▪ **LCM (Least Common Multiple):** Least common multiple of two or more given numbers is the 'least or lowest number' which is divisible by each of them exactly. In the sense without a non zero remainder.

Method of Finding LCM:

Step-1: Write numbers in a line separated by comma.

Step 2: Divide any two of the given numbers exactly with a least possible prime number then the quotients and the undivided numbers are written in the next line.

Step 3: Repeat the same process till all the numbers in the line are prime to each other i.e. no more common factors exist.

Conclusion: The product of all divisors and the numbers in the last line is the LCM of the numbers. e.g.: Find the LCM of 14, 18, 24, 30.

2	14, 18, 24, 30
3	7, 9, 12, 15
	7, 3, 4, 5

The LCM of 14, 18, 24, 30 = $2 \times 3 \times 7 \times 3 \times 4 \times 5 = 2520$. Observe another example below. **Square Root of 119716 is 346.**

Step 1: Group two digits as pairs. 11, 97, 16.

Step 2: Largest number whose square is near to the 11 is 3. Hence, 3 is the divisor and also quotient.

$$\begin{array}{r} 3 \overline{)11\ 97\ 16}\ (3 \\ \underline{9} \end{array}$$

Step 3: Now 297 is the new dividend.

$$\begin{array}{r} 3 \overline{)11\ 97\ 16}\ (3 \\ \underline{9} \\ 2\ 97 \end{array}$$

Step 4: Double the quotient 3 i.e. $3 \times 2 = 6$ and put a blank for a number beside 6 i.e. 6[?]. Now think of a largest number (for e.g., 4) to fill in the blank in such a way that the product of a new divisor (i.e. 64) and this digit (i.e. 4) is less than or equal to new dividend (i.e. 297).

Step 5:

$$\begin{array}{r} 3 \times 2 = 6 \\ \begin{array}{r} 3 \overline{)11\ 97\ 16}\ (3\ 4\ 6 \\ \underline{9} \\ 6\ 4 \overline{)2\ 97} \\ \underline{2\ 56} \\ 68\ 6 \overline{)41\ 16} \\ \underline{41\ 16} \\ 0 \end{array} \end{array} \quad \begin{array}{l} \downarrow \\ 34 \times 2 = 68 \end{array}$$

For this type of questions, it is better to check from options in the exam.

• Key Points on Finding Square Root:

1. A number ending with 2, 3, 7, 8 cannot be a perfect square. The last digit of any perfect square must be any one among 0, 1, 4, 5, 6, 9.
2. A number ending with odd number of zeros can never be a perfect square. e.g.: 1000, 2000 etc.
3. The difference between squares of two consecutive numbers is always an odd number. e.g.: $4^2 - 3^2 = 16 - 9 = 7$ (odd).

Finding square root of a decimal fraction:

First eliminate the decimal point by dividing and multiplying with even powers of 10 then find the square root of both numerator and denominator separately and then you can conclude the square root. e.g.: Find the square root of 1190.25.

$$\sqrt{\frac{1190.25}{10^2} \times 10^2} = \sqrt{\frac{119025}{10^2}} = \frac{345}{10} = 34.5$$

• Simplification:

In simplification we are supposed to follow the order which is essentially demanded by Mathematics and given by a common note of remembrance as VBODMAS. V – Vinculum ($\bar{x}$), B – Bracket () { }, O – of D – Division ($\div$), M – Multiplication ($\times$), A – Addition (+), S – Subtraction ($-$).

Use of Algebraic Identities:

The following algebraic identities will be useful in simplification.

1. $(a + b)^2 = a^2 + 2ab + b^2$
2. $(a - b)^2 = a^2 - 2ab + b^2$
3. $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$
4. $(a + b)^2 - (a - b)^2 = 4ab$
5. $a^2 - b^2 = (a + b)(a - b)$
6. $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$
7. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$

$$8. a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$9. a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$10. (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$11. a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$12. a^4 - b^4 = (a^2 + b^2)(a^2 - b^2)$$

• **Number of Divisors of a Composite Number:** If N is a composite number of the form $N = a^p b^q c^r \dots$ where a, b, c are primes, then the number of divisors of N is given by $(p+1)(q+1)(r+1) \dots$ e.g.: Let the number be 600.

$$\begin{array}{r} 2 \overline{) 600} \\ 2 \overline{) 300} \\ 2 \overline{) 150} \\ 2 \overline{) 75} \\ 2 \overline{) 25} \\ 5 \end{array}$$

$$600 = 2^3 \times 3^1 \times 5^2$$

$\therefore$ Number of divisors of 600 = $(3+1)(1+1)(2+1) = 24$. In these 24 divisors 1 and the number itself are also included. So, number of divisors of 600 excluding 1 and its self is $24 - 2 = 22$.

• **Sum of Divisors of a Composite Number :** If N is a composite number of the form $a^p b^q c^r \dots$ Where a, b, c are primes, then the sum of the divisors, S_N is given by $S_N =$

$$\frac{(a^{p+1} - 1)(b^{q+1} - 1)(c^{r+1} - 1)}{(a - 1)(b - 1)(c - 1)}$$

e.g.: Let the number be 600.

$$600 = 2^3 \times 3^1 \times 5^2$$

$$\begin{aligned} \text{Sum of the divisors } S_N &= \frac{(2^{3+1} - 1)(3^{1+1} - 1)(5^{2+1} - 1)}{(2 - 1)(3 - 1)(5 - 1)} \\ \Rightarrow \frac{(16 - 1)(9 - 1)(125 - 1)}{(1)(2)(4)} &\Rightarrow \frac{(15) \times (8) \times (124)}{(1) \times (2) \times (4)} = 1860 \end{aligned}$$

• **Important Key Points:**

1) Sum of natural numbers from 1 to $n = \frac{n(n+1)}{2}$.

2) Sum of squares of first n natural numbers $= \frac{n(n+1)(2n+1)}{6}$.

3) Sum of cubes of first n natural numbers $= \left[\frac{n(n+1)}{6} \right]^2$.

4) Number of odd numbers from 1 to $n = \frac{\text{Last Odd number} + 1}{2}$.

5) Number of even numbers from 1 to $n = \frac{\text{Last Even number}}{2}$.

6) Sum of even numbers from 1 to n is $k(k+1)$, where k indicates number of even numbers from 1 to n . e.g.: Sum of even no from 1 to 80 $= 40(40+1) = 1640$. Here from 1 to 80 there exists 40 even numbers.

7) Sum of odd numbers from 1 to $n = k^2$, where k is equal to number of odd numbers from 1 to n . e.g.: Sum of odd numbers from 1 to 60 is $(30)^2 = 900$. $(30)^2 = 900$. 30 odd natural numbers exist from 1 to 60.

8) Sum of the squares of first ' n ' even natural numbers $= \frac{2}{3} (n)(n+1)(2n+1)$.

9) Sum of squares of first ' n ' odd natural numbers is $\frac{n(2n+1)(2n-1)}{3}$.

10) Sum of any 5 consecutive whole numbers will always be divisible by 5.

e.g.: $(3 + 4 + 5 + 6 + 7) = 25$ is divisible by 5.

11) $XY - YX$; The difference between a two digit number and its reverse is divisible 9. e.g.: Let the two numbers be 95 and 59. Here 59 is reverse of 95. Now $95 - 59 = 36$ (which is divisible by 9).

12) Products: $\text{odd} \times \text{odd} = \text{odd}$;
 $\text{odd} \times \text{even} = \text{even}$;
 $\text{even} \times \text{even} = \text{even}$;

13) $n! = n(n-1)(n-2)(n-3) \dots (3)(2)(1)$. e.g.: $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$. Product of 'r' consecutive integers is divisible by (r!)

14) Finding the units digit of the numbers like $(252)^{54}$. Here the units digit of 252 is 2 and the index is 54. We know that $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $2^5 = 32$. Here units digit is repeated after each 4 indices. So divide 54 by 4 to get the remainder. Here remainder is 2. Hence the last digit in $(252)^{54}$ is same as 2^2 i.e. 4.

Sample Questions

1) The HCF of two numbers is 16 and their LCM is 160. If one number is 32, then other number is -

- a) 48 b) 80 c) 96 d) 112

2) Each student in a class contributed as many paise as the number of students in the class, the teacher contributed 13, the total collection is of 49. How many students were there in the class?

- a) 48 b) 60 c) 72 d) None

3) A biscuit dealer has 378 kgs, 434 kgs and 582 kgs of three different qualities of biscuits. He wants it all to be packed into boxes of equal size without mixing. Find the capacity of the largest possible box.

- a) 5 kg b) 3 kg c) 2 kg d) 1 kg

4) Two numbers 2035 and 2880 when divided by a certain number of three digits, leaves the same remainder. Find the number.

- a) 271 b) 169 c) 293 d) 421

5) Sum of 4 consecutive natural numbers each divisible by 5 is 130. What is the greatest number?

- a) 35 b) 40 c) 45 d) 50

6) The least perfect square number which is exactly divisible by 4, 6, 8, 10 or 12 is

- a) 9260 b) 7921 c) 5625 d) 3600

7) The last digit of a number 49825# is missing. It is also given that the number is divisible by 8. Find the digit at unit's place.

- a) 4 b) 6 c) 8 d) 0

8) If a boy saves 1 on day -1, 2 on day -2, 3 on day -3 and so on. Then in how many days will he have 36?

- a) 5 b) 6 c) 7 d) 8

9) Find the size of the largest square slabs which can be paved on the floor of a room 5 meters 44 cm long and 3 meters 74 cm broad.

- a) 56 b) 42 c) 38 d) 34

10) Find the size of the largest square slabs which can be paved on the floor of a room 5 meters 44 cm long and 3 meters 74 cm broad.

- a) 56 b) 42 c) 38 d) 34

11) Find the greatest number that will divide 148, 246 and 623 leaving remainders 4, 6 and 11 respectively.

- a) 11 b) 12 c) 13 d) 14

12) Find the least number which when divided by 36, 48 and 64 leaves the remainders 25, 37 and 53 respectively

- a) 656 b) 563 c) 565 d) 657

13) Three bells ring at an interval of 10, 12 and 14 seconds respectively. They ring together at 11:00 then at what time they ring together again.

- a) 12 hours 12 min 12 sec b) 11 hours 7 min
c) 11 hours 35 min d) 10 hours 45 min

14) The smallest number on being divided by 3, 4, 6, 10 and 16 leaves 9 as remainder in each case but is completely divisible by 9. What is that number?

- a) 720 b) 729 c) 846 d) None

15) Find the greatest number that will divide 532, 894 and 1003 leaving remainders 22, 44 and 68 respectively.

- a) 85 b) 105 c) 90 d) 95

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) B | 2) B | 3) C | 4) B |
| 5) B | 6) D | 7) C | 8) D |
| 9) D | 10) D | 11) B | 12) C |
| 13) B | 14) B | 15) A | |

Averages

$$\text{Average/Mean} = \frac{\text{Sum of } n \text{ observations}}{n} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

$$\text{Geometric Mean} = \sqrt[n]{X_1 \cdot X_2 \cdot X_3 \cdots X_n}$$

$$\text{Average Speed} = \frac{\text{Total Distance Travelled}}{\text{Total Time Taken}}$$

If equal distances are travelled at different speeds x_1 km/h, ..., x_n

$$\text{km/h, the average speed is given by } \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} \text{ km/h}$$

If the distance travelled at different speeds are not equal, the

$$\text{average speed is given by } \frac{d_1 + d_2 + \dots + d_n}{\frac{d_1}{x_1} + \frac{d_2}{x_2} + \dots + \frac{d_n}{x_n}} \text{ km/h}$$

If each observation is increased, decreased, multiplied or divided by any constant c , the average is also increased, decreased, multiplied or divided by c respectively.

Let the average of n quantities be equal to $\bar{x}$, then after removing a quantity the average $\bar{x}_1$. The value removed of the quantity is given by $[(\bar{x} - \bar{x}_1) + \bar{x}_1]$.

In the above case if a new quantity is added, the value of the new quantity is given by $[(\bar{x} - \bar{x}_1) + \bar{x}_1]$.

$$\text{The average of odd numbers from 1 to } n \text{ is } \left[\frac{\text{Last odd number} + 1}{2} \right]$$

$$\text{The average of even numbers from 1 to } n \text{ is } \left[\frac{\text{Last odd number} + 2}{2} \right]$$

Ex. In a class of 52 students the number of boys is two less than the number of girls. Average weight of boy is 42 kg, while the average weight of all the 52 students is 40 kg, Approximately what is the average weight of the girls?

- (A) 37 kg (B) 38 kg (C) 39 kg (D) 39.5 kg

Sol. Let the number of girls be x , then the number of boys is $(x-2)$.

$$x(x-2) = 52 \Rightarrow x = 27$$

$\therefore$ There are 27 girls and 25 boys in the class.

Average weight of boys = 42 kg

$$\therefore \text{Sum of their weights of boys} = 25 \times 42 = 1050 \text{ kg}$$

$$\text{Similarly, sum of the weights of the girls} = 52 \times 40 = 2080$$

$$\Rightarrow \text{Sum of the weights of the girls} = 2080 - 1050 = 1030 \text{ kg}$$

$$\therefore \text{Their average weight} = \frac{1030}{27} = 38.14 \text{ kg}$$

$$= 38 \text{ kg (approximately) i.e., (2)}$$

Ex. The average of Suresh's marks in English and History is 55. His average of marks in English and Science is 65. What is the difference between the marks which he obtained in History and Science?

- (A) 40 (B) 60 (C) 20 (D) NOTA

Sol. By given condition,

$$\frac{\text{English} + \text{History}}{2} = 55$$

$$\Rightarrow \text{English} + \text{History} = 110 \quad \dots (1)$$

$$\text{Similarly, English} + \text{Science} = 130 \quad \dots (2)$$

Ex. The average weight of 8 persons increase by 1.5 kg when a person weighting 65 kg is replaced by a new person, what could be the weight of the new person?

- (A) 76 kg (B) 77 kg (C) 76.5 kg (D) 77.5 kg

Sol. Let the average weight of 8 persons be $\bar{x}$ and the weight of the new person be w kg.

$$\text{Sum of weights of 8 persons} = 8$$

Sol. By the given condition, new sum = $\frac{8\bar{x} - 65 + w}{8} = \bar{x} + 1.5$

$$\Rightarrow 8\bar{x} - 65 + w = 8\bar{x} + 8 \Psi 1.5$$

$$\Rightarrow 2 = 12 + 65 = 77 \text{ kg}$$

Shortcut: Weight of new person = number of persons Ψ increment in average + weight of the person replaced
 $= 8 \Psi 1.5 + 65 = 77 \text{ kg.}$, i.e., (b)

Ex. The average of four consecutive even numbers is one fourth of the sum of these numbers. What is the difference between the first and the last number?

- (A) 4 (B) 6 (C) 2 (D) 5

Sol. Since the numbers are consecutive even integers, therefore they can be $x, x+2, x+4$ and $x+6$. Clearly, the difference between the first and last is $(x + 6) - x = 6$, i.e., (2)

Ex. Present ages of P and Q are in the ratio 5:6 respectively. Seven years hence this ratio becomes 6:7 respectively. What is P's present age in years?

- (A) 35 (B) 42 (C) 49 (D) NOTA

Sol. Let the ages of P and Q be $5x$ and $6x$ respectively.

By the given conditions,

$$5x + 7 : 6x + 7 = 6 : 7$$

$$\Rightarrow 7(5x+7) = 6(6x+7) \Rightarrow x = 7$$

$$\therefore \text{P's present age} = 5x = 35 \text{ years i.e., (1)}$$

Sample Questions

1) The average of 9 numbers is 30. The average of first 5 numbers is 25 and that of the last 3 numbers is 35. What is the 6th number?

- a) 20 b) 30 c) 40 d) 50

2) The arithmetic mean weight of 20 students in a class is increased by 0.75 kg when one of the students weighing 30 kg is replaced by a new student. Weight of new student (in kg) is

- a) 35 b) 40 c) 45 d) 50

3) The average marks scored by the students of a class is 68. The average marks of the girls in the class is 80 and that of the boy is

60. What is the percentage of boy in the class?

- a) 48 b) 80 c) 96 d) 112

4) The arithmetic mean of the scores of a group of students in a test was 52. The brightest 20% of them secured a mean score of 80 and the duller 25% a mean score of 31. The mean score of remaining 55% is

- a) 45% b) 50% c) 51.4% approx d) 54.6 % approx

5) The average of marks of 14 students was calculated as 71. But it was later found that the marks of one student had been wrongly entered as 42 instead of 56 and of another as 74 instead of 56 and of another as 74 instead of 32. The correct average is

- a) 67 b) 62 c) 69 d) 71

6) A father said to his son "I was as old as you are at present at the time of your birth". If the father's age is 38 years now, the son's age five years back was

- a) 14 b) 19 c) 18 d) 13

7) Out of the three numbers, the first is twice the second and is half of the third. If the average of the three numbers is 56, then difference of first and third numbers is

- a) 12 b) 20 c) 24 d) 48

8) Harsha is 40 years old and Rekha is 60 years old. How many years ago was the ratio of their ages 3 : 5?

- a) 10 b) 20 c) 37 d) 5

9) Sumitra has an average of 56% on her first 7 examinations. How much she should make on her eighth examination to obtain an average of 60% on 8 examinations?

- a) 88% b) 78% c) 98% d) None

10) Ages of 'A' and 'B' are in the ratio of 2 : 3 respectively. Six years later the ratio of their ages will become 8 : 11 respectively. What is B's present age?

- a) 48 b) 80 c) 96 d) 112

11) The total age of A and B is 12 years more than of total age of B and C. So, C is how many years younger than A?

- a) C is elder b) 24 c) 12 d) 16

12) The average marks fetched by Mohan in History, Geography, Science and Mathematics is 10 more than the marks fetched in Mathematics. If he has got 110 marks aggregate in History and Geography. What will be aggregate marks fetched in Science and Mathematics?

- a) 80 b) 70 c) 65 d) 60

13) Jayesh is twice as old as Vijay and half as old as Suresh. If sum of Suresh's and Vijay's ages is 85 years. What's Jayesh's age?

- a) 34 b) 36 c) 38 d) 40

14) The average weight of a group of 150 students in a class is 60 kg. If the mean of weights of all the 50 male students in the class is 70 kg, then the average weight of 100 girls in the class is (kg)

- a) 65 b) 50 c) 60 d) 55

15) Jitu scores 60, 80, 40, 50, and 90 percent marks in the subjects A, B, C, D and E respectively. However, the weights attached to those subjects are 5, 4, 3, 2 and 1 respectively. Which of the following represents the weight average of marks scored by Jitu in these five subjects?

- a) 60 b) 72 c) 62 d) 74

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) C | 2) C | 3) D | 4) C |
| 5) C | 6) A | 7) C | 8) A |
| 9) A | 10) A | 11) C | 12) C |
| 13) A | 14) D | 15) C | |

Percentages

If an object's price is increased or decreased by $x\%$ and the factors are increased or decreased by $y\%$ the net effect is given by

$$\text{Net Effect} = \left[x + y + \frac{xy}{100} \right] \%$$

Remember that the percentages are taken with positive or negative sign according as there is increase or decrease in the factor.

If the net effect is nil, i.e., there is no loss or gain, then the above formula becomes

$$y = \left[\frac{100x}{100 + x} \right]$$

If A's income is $x\%$ more than that of B, B's income is less than that of A by

$$\left[\frac{100 \cdot x}{100 + x} \right] \%$$

If A's income is $x\%$ less than that of B, B's income is more than that of A by

$$\left[\frac{100 \cdot x}{100 - x} \right] \%$$

Population Formulae :

1. If the original (present) population of a town is P , then the population (p_0) after n years at annual increase of $r\%$ is given by

$$P_0 = P \left(1 + \frac{r}{100} \right)^n$$

2. If the present population is P , then the population n years ago is given by

$$P_0 = \frac{P}{\left(1 + \frac{r}{100} \right)^n}$$

If the population increased by $x\%$ during the first year, by $y\%$ during the second year, by $z\%$ during the third year, the population after three years will be

$$P \left(1 + \frac{x}{100}\right) \left(1 + \frac{y}{100}\right) \left(1 + \frac{z}{100}\right)$$

Depreciation Formulae: (1) If the present price of an article is P , then the price P_0 after n years at an annual decrease of $r\%$ is given by

$$P_0 = P \left(1 - \frac{r}{100}\right)^n$$

(2) If the present price is P , then the price n years ago is given by

$$P_0 = \frac{P}{\left(1 - \frac{r}{100}\right)^n}$$

If after spending $p_1\%$ first, then $p_1\%$ from the remaining, and so on, B is the balance amount, then the total (original) amount is given by

$$\text{Total Amount} = \frac{B \times 100 \times 100}{(100 - p_1)(100 - p_2) \dots}$$

If the value of a number is first increased by $x\%$ and later decreased by $x\%$, the net change is always a x^2 decrease which is equal to $\frac{x^2}{100}\%$.

A candidate scoring $x\%$ in an examination fails by 'a' marks, while another candidate who scores $y\%$ marks gets b' marks more than the minimum required pass marks. Then the maximum marks for

that examination are $M = \frac{100(a + b)}{y - x}$

If in an examination $x\%$ of the students failed in one subject, $y\%$ failed in another subject and 29% failed in both the subjects, the percentage of student who:

- (i) failed in either or the subjects = $x + y - z$
(ii) passed in both the subjects = $100 - (x + y - z)$

If the price of an article is successively increased by $x\%$, $y\%$ and $z\%$, then single equivalent increase in the price will be

$$\left[x + y + z + \frac{xy + yz + zx}{(100)} + \frac{xyz}{(100)^2} \right] \%$$

Sample Questions

- 1) In an examination, a student who gets 20% of the maximum marks fails by 5 marks. Another student who scores 30% of the maximum marks get 20 marks more than pass marks. The necessary percentage required for passing is
a) 32% b) 23% c) 22% d) 20%
- 2) In a class, 60% of the students pass in Hindi and 45% pass in Sanskrit. If 75% of them pass in at least one subject, what is the percentage of the student fail both in these subjects?
a) 80% b) 75% c) 20% d) 25%
- 3) If A's income is 40% less than that of B, how much percent B's income is more than that of A?
a) 60% b) 40% c) 66.66% d) 33.33%
- 4) In an examination 80% students passed in Mathematics and 75% passed in English, while 15% of the students failed in both the subjects. Find the total number of students who appeared in the examination.
a) 500 b) 450 c) 540 d) 550
- 5) An ore contains 26% copper. To get 91 kg of copper the quantity of the ore required is
a) 350 kg b) 250 kg c) 240 kg d) 450 kg
- 6) The length of rectangle is doubled while it's breadth is halved. What is the percentage change in area?
a) 25 b) 55 c) 75 d) None
- 7) The price of petrol went up by 25%. In order that expenses on petrol should not increase, one must reduce travel by

- a) 25% b) 20% c) 18% d) 15%

8) The population of a town is 12000. If the male population is increased by 6% every year and that of female by 8% then the total population at the end of the year becomes 12800. Find the number of males in the town.

- a) 8000 b) 7500 c) 7200 d) 6500

9) A fruit seller has certain number of apples of which 5% are rotten. He sells 93% of the remainder and then has 266 left. How many apples he had originally?

- a) 1000 b) 2000 c) 3000 d) 4000

10) A screwdriver and a hammer cost the same now. If the price of a screwdriver increases by 5% and that of the hammer by 3%, how much more will it cost to buy 3 screwdrivers and 3 hammers?

- a) 4% b) 5% c) 8% d) 24%

11) If 8% of x is the same as 4% of y , then 20% of x is the same as

- a) 10% of y b) 16% of y c) 80% of y d) 50% of y

12) If a man buys an article at $\frac{3}{4}$ th of its value and sells it for 50% more than its value, what is his percent profit based on?

- a) 50 b) 100 c) 75 d) 125

13) A 7 litre solution of water contains 20% of a chemical "Xylene" should be added to the solution in order to increase its concentration level to 30%?

- a) 1 litre b) 1.4 litre c) 2 litre d) 2.5 litre

14) In order to increase the total revenue, a trader announced 25% discount. As a result, the sales volume, increased by 40%. What is the net effect on the sales revenue?

- a) 15% increase b) 15% decrease c) 5% increase d) 5% decrease

15) A shopkeeper allows two successive discounts of 10% and 20%. If he sells an article for Rs. 540, then the marked price of the article is

- a) 750 b) 740 c) 725 d) 700

Solutions

1) C

2) D

3) C

4) A

5) A

6) D

7) B

8) A

9) D

10) A

11) A

12) B

13) A

14) C

15) A

Ratio Proportion

Ratio: A ratio is the relation between two quantities which is expressed by a fraction.

- The ratio of the number 'a' to the number 'b' is written as

$\frac{a}{b}$ (or) $a : b$ or a to b . e.g.: The ratio of 5 hours to 3 hours can be

written as $\frac{5}{3}$ (or) $5 : 3$.

- The ratio is always a comparison between the quantities of same kind or of same units. For example, you cannot form the ratio between 5 hours and 3 days. Because the two numbers are expressed in different units. Hence, convert 3 days to hours. i.e. 3 days = 72 hours. Thus the proper form of this ratio is

$\frac{5}{72}$ (or) $5 : 72$.

- Two quantities which are being compared ($a : b$) are called its terms. The first term (a) is called antecedent and second term (b) is called consequent.
- The ratio of two quantities is always an abstract number (without any units).
- If the terms of a ratio are multiplied or divided by the same quantity the value of the ratio remains unaltered. e.g.: The ratio $a : b$ is same as $Ma : Mb$.

Proportion: Equality of two ratios is called proportion. Consider the two ratios, $a : b$ and $c : d$, then proportion is written as, $a : b ::$

$c : d$ (or) $a : b = c : d$ (or) $\frac{a}{b} = \frac{c}{d}$ Here a, b, c, d are called Terms. $a,$

d are called Extremes (end terms) and b, c are called Means (middle terms).

e.g.: Since the ratio 4:20 (or) $\frac{4}{20}$ is equal to the

ratio 1:5 or $\frac{4}{20} = \frac{1}{5}$.

• In a proportion, product of means (middle terms) is equal to product of extremes (end terms). i.e. $ad=bc$ or $\frac{a}{b} = \frac{c}{d}$.

Key Notes: If a and b are two quantities, then

1) Duplicate ratio of $a : b = a^2 : b^2$

2) Sub-duplicate ratio of $a : b = \sqrt{a} : \sqrt{b}$

3) Triplicate ratio of $a : b = a^3 : b^3$

4) Sub-triplicate ratio $a : b = \sqrt[3]{a} : \sqrt[3]{b}$

5) Inverse or reciprocal ratio of $a : b = \frac{1^2}{a} : \frac{1^2}{b}$.

6) Third proportional of a and b is defined to be that number c such that $a : b = b : c \Rightarrow c = \frac{b^2}{a}$.

7) If $a : b = x : y$ and $b : c = p : q$, then

a) $a : c = \frac{x \times p}{y \times q}$.

b) $a : b : c = px : py : qy$

8) Compound Ratio of $(a : b)$, $(c : d)$, $(e : f)$ is $\frac{a}{b} \times \frac{c}{d} \times \frac{e}{f}$.

9) The ratio in which two kinds of substances must be mixed together one at ₹x per kg and another at ₹y per kg, so that the

mixture may cost ₹n per kg. The ratio is $\frac{n - y}{x - n}$.

10) Let the incomes of two persons be in the ratio of $a : b$ and their expenditure be in the ratio of $x : y$ and if the savings of each person is ₹ n then income of each is

₹ $\frac{an(y - x)}{ay - bx}$ and ₹ $\frac{bn(y - x)}{ay - bx}$ respectively.

11) In a mixture the ratio of milk and water is $a:b$. In this mixture another n liters of water is added, then the ratio of milk and water in the resulting mixture became $a : m$. Then, the quantity of milk

in the original mixture = $\frac{an}{m - b}$ and the quantity of water in the

original mixture = $\frac{bn}{m - b}$.

12) In a mixture of n liters, the ratio of milk and water is $x : y$. If another m liters of water is added to the mixture, the ratio of milk and water in the resulting mixture = $xn:(yn + mx + my)$.

13) If four numbers a, b, c and d are given then

a) $\frac{ad - bc}{(b + c) - (a + d)}$ should be added to each of these numbers so that the resulting numbers may be proportional.

b) $\frac{ad - bc}{(a + d) - (b + c)}$ should be subtracted from each of these numbers so that the resulting numbers may be proportional.

Sample Questions

1) The annual incomes of two persons are in the ratio 9: 7 and their expenses are in the ratio 4: 3. If each of them saves Rs. 2000 per year, then their annual incomes are:

- a) Rs. 18000 and Rs. 14000 b) Rs. 36000 and Rs. 28000
c) Rs. 27000 and Rs. 21000 d) Rs. 45000 and Rs. 35000

2) A person divided Rs. 10800 among his 3 sons in the ratio 3: 4: 5. Second son kept Rs. 1000 for himself, gave Rs. 600 to his wife, and divided the remaining money among his 2 daughters in the ratio 11: 9. Then one of his daughters received:

- a) 1000 b) 1050 c) 1100 d) 1150

3) In a school the number of boys and that of the girls are in the respective ratio of 2: 3. If the number of boys is increased by 20% and that of girls is increased by 10%, what will be the new ratio of number of boys to that of the girls?

- a) 14 : 5 b) 5 : 8 c) 13 : 4 d) None

4) Income of two companies A and B are in the ratio of 5 : 8. Had the income of company A been more by Rs. 25 lakhs, the ratio of their incomes would have been 5: 4 respectively. What is the income of company B?

- a) Rs. 80 lakhs b) Rs. 50 lakhs c) Rs. 40 lakhs d) Rs. 60 lakhs

5) A total of 91 boys are seated in three rows. The ratio between the number of boys seated in the first and the second row is 5 : 2 respectively and the ratio between the number of boys seated in the second and the third row is 1 : 3 respectively. How many boys were there in the second row?

- a) 35 b) 14 c) 42 d) 56

6) The ratio of the income of A and B as also those of B and C is 3 : 2. If a third of A's income exceeds a fourth of C's income by Rs. 1000, what is B's income?

- a) 2000 b) 3000 c) 2500 d) 3500

7) A sum of money is divided among 160 males and some females in the ratio 16: 21. Individually each male gets Rs. 4 and each female Rs. 3. The number of females is:

- a) 280 b) 198 c) 284 d) 270

8) The monthly income of two persons are in the ratio 4: 7, and their expenses are in the ratio 11:20. If each of them saves Rs. 400 per month, then their monthly income must be respectively:

- a) Rs. 3600, Rs. 4200 b) Rs. 4200, Rs. 7350
c) Rs. 4000, Rs. 7000 d) Rs. 4800, Rs. 8400

9) Divide Rs. 1250 among A, B and C so that A gets $\frac{2}{9}$ of B's share and C gets $\frac{3}{4}$ of A's share

- a) Rs. 200, Rs. 800, Rs. 250 b) Rs. 150, Rs. 800, Rs. 300

c) Rs. 100, Rs. 900, Rs. 250 d) Rs. 200, Rs. 900, Rs. 150

10) Rs. 180 contained in a box consists of one-rupee, 50-paise and 25-paise coins in the ratio 2: 3: 4. What is the number of 50-paise coins?

a) 60 b) 120 c) 150 d) 180

11) The students in three classes are in the ratio 2: 3: 5. If 20 students are increased in each class, the ratio changes to 4:5:7. The total number of students before the increase were

a) 10 b) 90 c) 95 d) None

12) If 378 coins consist of one rupee, 50-paise and 25-paise coins whose values are in the ratio of

a) 132 b) 128 c) 136 d) 133

13) The ratio of the number of boys to that of girls in a school is 4 : 1. If 75% of boy and 70% of the girls are scholarship holders, then the percentage of students who do not get scholarship is:

a) 1 litre b) 1.4 litre c) 2 litre d) 2.5 litre

14) In a factory men, women and children were employed in the ratio 8: 5: 1 to finish a job and their individual wages were in the ratio 5: 2 : 3. Total daily wages of all amount of Rs. 318. Find the total daily wages paid to each category.

a) Rs. 240, 60, 20 b) Rs. 210, 70, 38
c) Rs. 190, 95, 33 d) None

15) A total profit of Rs. 3600 is to be distributed amongst A, B and C such that A: B: 5: 4 and B:C: 8:9. What is the share of C?

a) Rs. 1200 b) Rs. 1800 c) Rs. 1500 d) Rs. 1350

Solutions

1) A	2) C	3) D	4) C
5) B	6) B	7) A	8) D
9) D	10) B	11) D	12) A
13) D	14) D	15) A	

Chain Rule

Direct Proportion

A↑ B↑

A↓ B↓

When quantity A increases, quantity B also increases, or when A decreases, B also decreases.

For example: The cost of 8 apples is Rs. 80. If the number of apples increases, the total cost will also increase.

Solving:

Step 1: Divide

Step 2: Multiply

1. The area occupied by 10 postage stamps is 50 cm². Find the total area occupied by 100 such postage stamps.

Solution: This is a case of direct proportion, so we will first divide and then multiply.

Step 1: $50 / 10 = 5$

Step 2: $5 \times 100 = 500\text{cm}^2$

Inverse Proportion

A↑ B↓

A↓ B↑

When quantity A increases, quantity B decreases, and vice versa.

For example: If you increase your speed, the time taken to reach a destination will decrease. Similarly, if more people work together on a task, the time required to complete the task will reduce.

Solving:

Step 1: Divide

Step 2: Multiply

2. If 35 men can complete a piece of work in 8 days, how many days will it take 20 men to complete the same work?

Solution: This is a question of inverse proportion.

Step 1: 35×8

Step 2: $(35 \times 8) / 20$
= 14 Days

Work done = Number of persons $\times$ Time (Day $\times$ Hours)

$W = M \times T$

Here the unit of work is man days or man hours. The problems of time and work can primarily be divided into two types.

1. The first type includes problems where individuals work with different efficiencies, alone or in combination, to complete a task.
2. The second type includes problems where group efficiencies are involved. Chain Rule concepts can be applied to group efficiency problems.

Also, when two work scenarios are given, we can change the above formula as -

$$\frac{(M1D1T1)}{W1} = \frac{(M2D2T2)}{W2}$$

Where

- M1** - Number of men in the 1st scenario,
- D1** - Number of days in the 1st scenario,
- T1** - Number of hours worked per day in the 1st scenario,
- W1** - Total work completed in the 1st scenario.
- M2** - Number of men in the 2nd scenario,
- D2** - Number of days in the 2nd scenario,
- T2** - Number of hours worked per day in the 2nd scenario,
- W2** - Total work completed in the 2nd scenario.

Sample Questions

1) If 30 laborer's working 7 hours a day can finish a piece of work in 18 days, how many laborer's working 6 hours a day can finish it in 30 days

a) 15

b) 21

c) 22

d) 25

2) 4 mat-weavers can weave 4 mats in 4 days. At the same rate, how many mats would be woven by 8 mat-weavers in 8 days?

- a) 4 b) 8 c) 12 d) 16

3) A certain number of persons can dig a trench 100 m long, 50 m broad and 10 m deep in 10 days. The same number of persons can dig another trench 20 m broad and 15 m deep in 30 days. The length of the second trench is :

- a) 400 b) 800 c) 500 d) 600

4) If 18 pumps can raise 2170 tonnes of water in 10 days, working 7 hours a day, in how many days will 16 pumps raise 1736 tonnes, working 9 hours a day?

- a) 7 b) 8 c) 4 d) 10

5) A garrison of 500 men had provisions for 27 days. After 3 days a reinforcement of 300 men arrived. For how many more days will the remaining food last now?

- a) 15 b) 16 c) 17.5 d) 18

6) A wheel that has 12 cogs is meshed with a larger wheels of 24 cogs. When the smaller wheel has made 40 revolutions, the number made by the larger wheel is:

- a) 14 b) 20 c) 12 d) 30

7) In a camp, there is a meal for 120 men or 200 children. If 150 children have taken the meal, how many men will be catered to with the remaining meal ?

- a) 20 b) 30 c) 40 d) 50

8) A certain number of men can finish a piece of work in 100 days. If there were 10 men less, it would take 10 days more for the work to be finished. How many men were there originally?

- a) 110 b) 70 c) 90 d) 100

9) A contractor undertook to do a certain piece of work in 18 days. He employed certain number of men but 12 of them being absent from the very 1st day, the rest could finish the work in 30 days. Find the number of men originally employed

- a) 20 b) 25 c) 27 d) 30

10) A contractor undertakes to do a piece of work in 40 days. He engages 100 men at the beginning and 100 more after 35 days and completes the work in stipulated time. If he had not engaged the additional men, how many days behind schedule would it be finished?

- a) 5 b) 6 c) 7 d) 8

11) A contractor employed 30 men to complete the project in 100 days. But later on he realized that just after 25 days only 20% of the work had been completed.- How many extra days, than the scheduled time are required?

- a) 25 b) 30 c) 22 d) 28

12) If 5 engines consume 6 metric tonnes of coal when each is running 9 hours a day, how much coal (in metric tonnes) will be needed for 8 engines, each running 10 hours a day, it being given that 3 engines of the former type consume as much as 4 engines of latter type?

- a) 10 b) 8 c) 9 d) 15

13) If 6 men and 8 boys can do a piece of work in 10 days and 26 men and 48 boys can do the same in 2 days, the time taken by 15 men and 20 boys to do the same type of work will be :

- a) 5 days b) 4 days c) 6 days d) 7 days

14) 5 men and 5 women earn Rs. 660 in 3 days. 10 men and 20 women earn Rs. 3500 in 5 days. In how many days can 6 men and 4 women earn Rs. 1060?

- a) 5 days b) 4 days c) 6 days d) 7 days

15) 5 men or 8 women do equal amount of work in a day. A job requires 3 men and 5 women to finish the job in 10 days how many woman are required to finish the job in 14 days?

- a) 5 b) 9 c) 7 d) 10

Solutions

- | | | | |
|------|------|------|------|
| 1) B | 2) D | 3) C | 4) A |
| 5) A | 6) B | 7) B | 8) A |

9) C

10) A

11) A

12) B

13) B

14) A

15) C

Profit, Loss and Discounts

Cost Price (CP) is the price at which article is bought.

Selling Price (SP) is the price at which article is sold.

Marked Price (MP) or List Price is the price marked on the article. For example, a vendor buys 1kg of mangoes for ₹50. He labeled the price as 80. But sold ₹ for 70. ₹ Here CP = ₹50, MP = 80, ₹SP = ₹70. The expenses incurred on transportation, maintenance, packaging, advertisement etc. are considered as overhead. These overheads and the profit when added to the cost price determine the selling price.

Profit or Gain: Profit is made when the selling price is greater than the cost price.

$$\text{Profit} = \text{SP} - \text{CP}; \quad \text{Profit}\% = \frac{\text{Profit}}{\text{Cost Price}} \times 100$$

Considering the same example given above, Profit = 70 - 50 = 20.
₹ Profit % = $\frac{20}{50} \times 100 = 40\%$

Loss: Loss is made when the cost price is greater than the selling price.

$$\text{Loss} = \text{CP} - \text{SP}; \quad \text{Loss}\% = \frac{\text{Loss}}{\text{Cost Price}} \times 100$$

- Profit or Loss is calculated on cost price only. Discount is always calculated on the marked price.

$$\text{Discount} = \text{MP} - \text{SP}; \quad \text{Discount}\% = \frac{\text{Loss}}{\text{Cost Price}} \times 100$$

Consider the same example given above, Discount = 80 - 70 = 10;
Discount% = $\frac{10}{80} \times 100 = 12.5\%$

- To calculate Gain, Loss, Selling Price and Cost Price directly use the formula, Use + sign for profit and – sign for loss.

$$\text{SP} = \frac{(100 \pm \text{Gain or Loss}) \times \text{CP}}{100}$$

Example: Cost Price of an article is 70. At what price it ₹ should be sold in order to gain 20%?

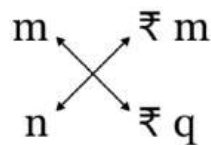
$$SP = (100 + 20) \times 70$$

$$100 = 120 \times 70$$

$$100 = 12 \times 7 = 84$$

$$SP = \frac{(100 + 20) \times 70}{100} = \frac{120 \times 70}{100} = 12 \times 7 = 84$$

• If a man purchased m articles for ₹ p and sold n articles for ₹ q . Then how much profit or loss does he make?



$$\text{Profit or Loss\%} = \frac{mp - np}{np} \times 100$$

Example: A merchant purchased 7 watches for ₹500 and sold 5 watches for ₹400. What is loss/gain percent?

$$\text{Explanation: } \frac{7 \times 400 - 5 \times 500}{5 \times 500} \times 100 = \frac{2800 - 2500}{2500} \times 100 = \frac{300}{25} = 12$$

• By selling an article for ₹ P , a merchant would gain or loss $x\%$. The price at which he sell it to gain or loss $y\%$ is

$$SP = P \left(\frac{100 \pm y}{100 \pm x} \right). \text{ (+ sign for gain; - sign for loss)}$$

Example: By selling a furniture for 180 a merchant ₹ will loss 10%. At what price must he sell to gain 20%.

$$\text{Explanation: } SP = 180 \times \left(\frac{100 + 20}{100 - 10} \right) = 240$$

• When a man buys two things on equal price each and in those things one is sold at a profit of $x\%$ and another is sold at a loss of $x\%$, then there will be no loss or no gain percent. Example: A merchant purchased a watch and a bag for 100 each. But he sold the watch at a profit of 20% and ₹ bag at a loss of 20%. What is his loss/gain percentage.

Example: A merchant purchased a watch and a bag for 100 each. But he sold the watch at a profit of 20% and ₹ bag at a loss of 20%. What is his loss/gain percentage?

Explanation: **CP** **SP**

Watch - ₹100 + 20% profit = ₹120

$$\text{Bag} - ₹100 - 20\% \text{ loss} = ₹80$$

₹200

₹200

Cost price = Selling Price. Hence, no gain or no loss.

- By selling two articles at the same price a merchant incurs $x\%$ loss on the first article and $x\%$ gain on the second article. In such a case there is always a loss.

$$\text{Loss} = \frac{2 \times \text{SP}}{\left(\frac{100}{X}\right)^2 - 1}$$

Example: By selling a watch and a bag at 100 each a ₹ merchant incurred a loss of 20% on watch and gain of 20% on bag. What is his loss or gain percentage?

Explanation: **CP** **SP**

Watch - ₹100 (20% loss on CP) = ₹125

$$\text{Bag} - ₹100 (20\% \text{ profit on CP}) = ₹83.33$$

₹200

₹208.33

Here, CP > SP. Hence, Loss = $\frac{8.33}{208.33} \times 100 = 3.9\%$

(or) Using Formula:
$$\text{Loss} = \frac{2 \times 100}{\left(\frac{100}{20}\right)^2 - 1} = \frac{200}{24}$$

$$= 8.33.$$

Then Cost Price = $200 + 8.33 = 208.33$ And Loss Percentage =

$$\frac{8.33}{208.33} \times 100 = 3.9\%$$

Sample Questions

- 1) A person purchased a chair at Rs. 70 and spent Rs. 17 on its

repair and 50 paise on cartage. If he sold the chair at Rs. 100, then his approximate margin of profit was:

- a) 11.25% b) 12.50% c) 13.30% d) 14.30%

2) A hawker gains the selling price of 4 ball-point pens on selling 1 dozen pens. His gain percentage is

- a) 25 b) $100 / 3$ c) 50 d) $199 / 3$

3) A person incurs a loss of 6% by selling a watch at Rs. 423. In order to get a profit of 6%, the person should sell the watch at

- a) 486 b) 477 c) 586 d) 577

4) A reduction of 20% in the price of oranges enables a man to buy 5 oranges more for Rs. 10. The price of an orange before reduction was

- a) 20 paise b) 40 paise c) 50 paise d) 60 paise

5) If 3 toys are sold at the cost price of 4 toys of the same kind, the profit will be

- a) 25% b) 23.34% c) 66.67% d) None

6) A shopkeeper marks his goods 20% above cost price, but allows 30% discount for cash. His net loss is

- a) 8% b) 10% c) 16% d) 20%

7) The selling price of 5 articles is the same as the cost price of 3 articles. The gain or loss is:

- a) 20% gain b) 25% gain c) 33.33% loss d) None

8) By selling an article for Rs. 480 a person lost 20%. For what should he sell it to make a profit of 20%?

- a) 800 b) 760 c) 720 d) 680

9) A tradesman marks his goods 10% above his cost price. If he allows his customers 10% discount on the marked price, how much profit or loss does he make, if any?

- a) 1% gain b) 1% loss c) 5% loss d) 0

10) Oranges are bought at 7 for Rs. 3. At what rate per hundred must they be sold to gain 33%?

- a) Rs. 56 b) Rs. 60 c) Rs. 58 d) Rs. 57

11) Applied to a bill for Rs. 1,00,000 the difference between a discount of 40% and two successive discounts of 36% and 4% is

- a) Nil b) Rs. 1,440 c) Rs. 2,500 d) Rs. 4,000

12) A sells a bicycle to B at a profit of 20%. B sells it to C at a profit of 25%. If C pays Rs. 225 for it, the cost price of the bicycle for A is

- a) Rs. 110 b) Rs. 125 c) Rs. 120 d) Rs. 150

13) Twelve copies of a book were sold for Rs. 1800 there by gaining cost-price of 3 copies. The cost price of a copy is

- a) Rs. 120 b) Rs. 150 c) Rs. 1200 d) Rs. 1500

14) A trader bought 10 kg of apples for Rs. 405 out of which 1 kg of apples wishes to make a profit of 10%, at what rate should he sell the remaining apples per kg?

- a) Rs. 45 b) Rs. 49.50 c) Rs. 50 d) Rs. 51

15) Marked price of a watch was Rs. 720. A man bought the same for Rs. 550.80, after getting two successive discounts, the first at 10%. What was the second discount rate ?

- a) 12% b) 14% c) 15% d) 18%

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) D | 2) C | 3) B | 4) C |
| 5) D | 6) C | 7) D | 8) C |
| 9) B | 10) D | 11) B | 12) D |
| 13) A | 14) B | 15) C | |

Mixtures and Allegations

Compound Mixture

A compound mixture is produced when two or more simple mixture are combined.

Example: Coffee soda

Allegation

Allegation is a way of resolving mathematical problems involving element combinations. This rule allows us to determine the ratio in which two or more ingredients at the particular price must be blended to attain the desired price mixture.

Mean Price

The cost price of a unit quantity of a mixture made by combining two or more elements.

These concepts can be solved by the following methods

1. Formula Method
2. Criss-Cross Method

Formula Method

The general formula to find the ratio of ingredients in the mixture

$$\text{is: } \frac{\text{Quantity of Cheaper}}{\text{Quantity of Dearer}} = \frac{\text{Cost price of Dearer} - \text{Mean price}}{\text{Mean price} - \text{Cost price of the cheaper}}$$

Let's understand the formula method better with an example:

Example Problem

In what proportion must tea at Rs. 7.50 per kg be mixed with tea at Rs. 10.50 per kg to produce a mixture worth Rs. 8.50 per kg?

Solution: Applying the formula method with the given data, we

$$\text{get, } \frac{\text{Quantity of Cheaper}(X)}{\text{Quantity of Dearer}(Y)} =$$

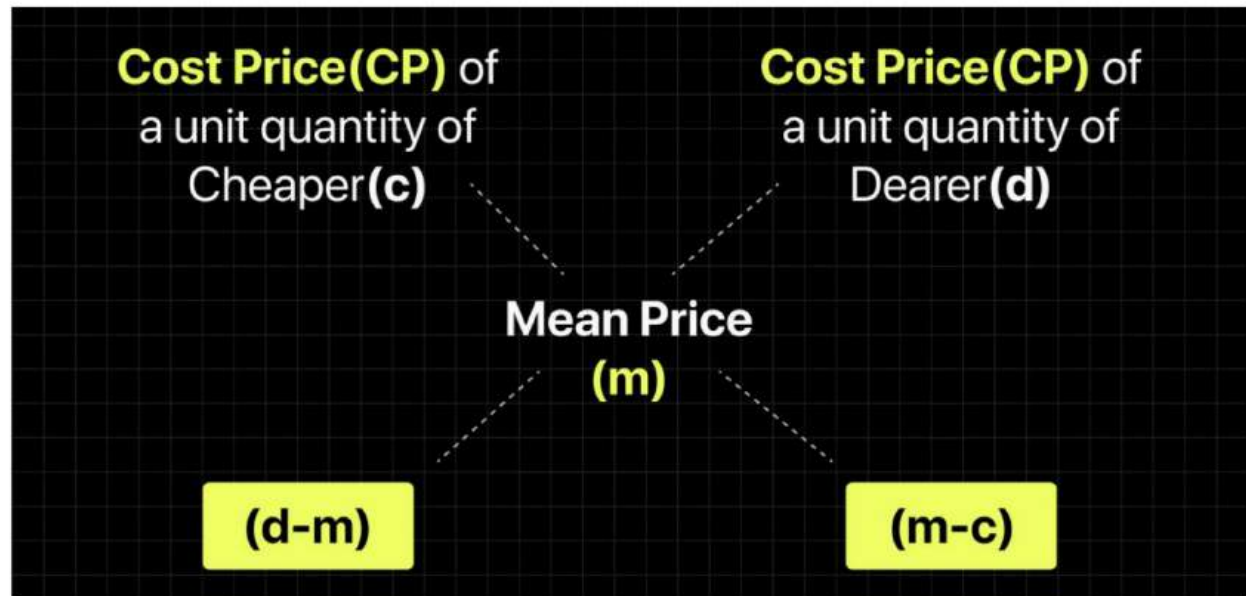
$$\frac{\text{Cost price of Dearer}(10.5 \text{ Rs}) - \text{Mean price}(8.5 \text{ Rs})}{\text{Mean price}(8.5 \text{ Rs}) - \text{Cost price of the cheaper}(7.5 \text{ Rs})}$$

$$\Rightarrow \frac{X}{Y} = \frac{10.5 - 8.5}{8.5 - 7.5} \Rightarrow \frac{X}{Y} = \frac{2}{1}$$

So we get the final ratio as 2 : 1.

Criss-Cross Method

The criss-cross method can be expressed using a diagram:



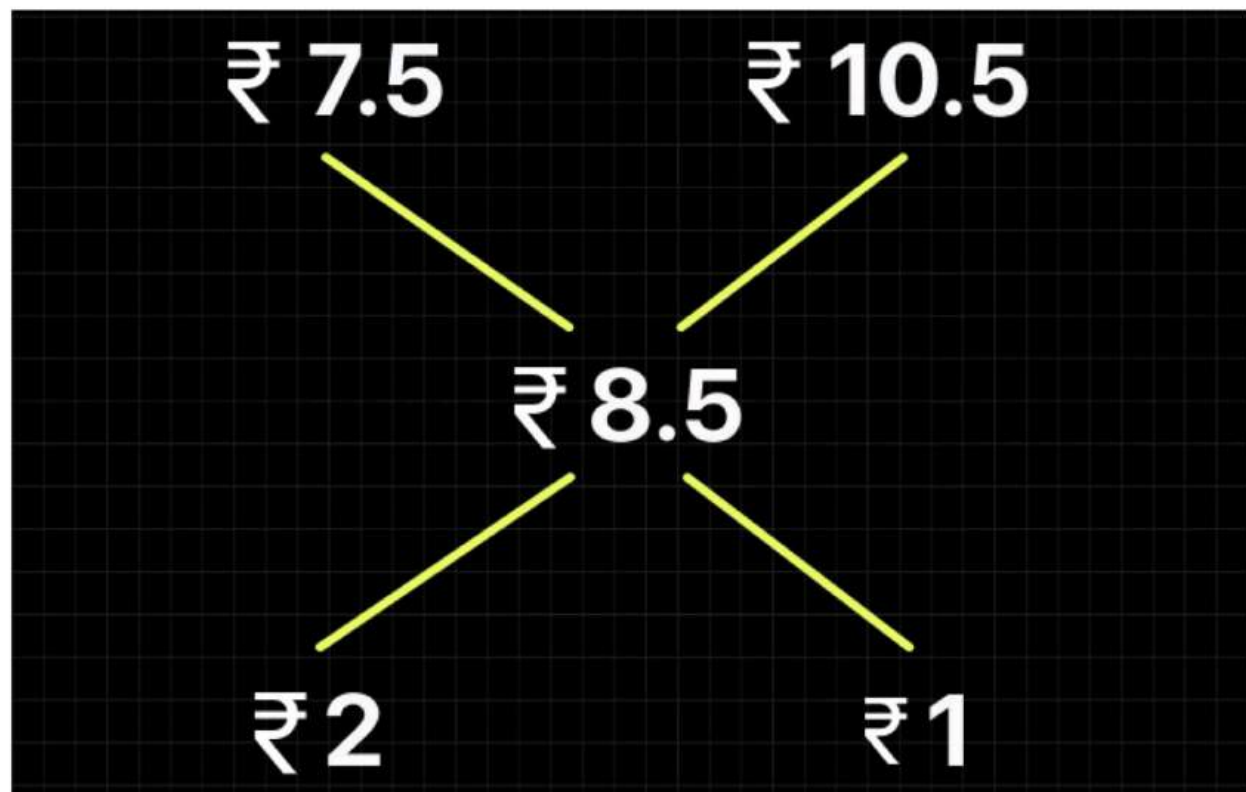
Let's understand the criss-cross method better with an example:

Example Problem

In what proportion must tea at Rs. 7.50 per kg be mixed with tea at Rs. 10.50 per kg to produce a mixture worth Rs. 8.50 per kg?

Solution

Using the criss-cross method, we get the answer as 2 : 1.



Sample Questions

1) A can contains a mixture of two liquid A and B in the ratio 7 : 5. When 9 litres of mixture are drawn off and can is filled with B, the ratio of A and B becomes 7 : 9. How many litres of liquid A was contained by the can initially?

- a) 25 b) 21 c) 20 d) 10

2) Prabodh bought 30 kg of rice at the rate of Rs. 8.50 per kg and 20 kg of rice at the rate of Rs. 9.00 per kg. He mixed the two. At what price (approx.) per kg should he sell the mixture in order to get 20% profit?

- a) Rs. 9.50 b) Rs. 8.50 c) Rs. 10.50 d) Rs. 12.00

3) Prabhu purchased 30 kg. He mixed the two and sold the entire quantity at the rate of Rs. 18.60 per kg. and made 20 percent overall. profit. At what price per kg. did he purchase the lot of another 30 kg. rice?

- a) Rs. 14.50 b) Rs. 12.50 c) Rs. 15.50 d) Rs. 13.50

4) Nikita bought 30 kg of wheat at the rate of Rs. 4.75 per kg, 40 kg of wheat at the rate of Rs. 4.25 per kg and mixed them. She sold the mixture at the rate of Rs. 4.45 per kg. Her total profit or loss in the transaction was

- a) Rs. 1 loss b) Rs. 1 profit c) Rs. 3.50 loss d) Rs. 3.50 profit

5) A mixture contains spirit and water in the ratio 3 : 2. If it contains 3 litres more spirit than water, the quantity of the spirit in the mixture is

- a) 10 litres b) 12 litres c) 8 litres d) 9 litres

6) A container contains 240 litres of wine, 80 litres is taken out of the container everyday and an equal quantity of water is put into it. Find the quantity of the wine that remains in the container at the end of the fourth day.

- a) 39.2 litres b) 32 litres c) 42.5 litres d) 47.40 litres

7) A vessel contains 50 litres milk. The milkman delivers 10 litres to the first house and adds an equal quantity of water. He does exactly the same at the second and third house. What is the

ratio of milk and water when he has finished delivering at the third house?

- a) 64 : 61 b) 27 : 37 c) 16 : 19 d) 19 : 27

8) Several litres of acid were drawn off from a 54 litres vessel full of acid and an equal amount of water is added. Again the same volume of the mixture was drawn off and replaced by water. As a result, the vessel contained 24 litres of pure acid. How much acid was drawn off initially ?

- a) 12 litres b) 16 litres c) 18 litres d) 24 litres

9) A mixture of a certain quantity of milk with 16 litres of water is worth 90 paise per litre. If pure milk be worth 108 paise per litre, how much milk in there is the mixture ?

- a) 60 litres b) 20 litres c) 80 litres d) 90 litres

10) A mixture of a certain quantity of milk with 32 litres of water is worth Rs. 1.50 per litre. If pure milk be worth Rs. 4.50 per litre, how much milk is there in the mixture?

- a) 18 litres b) 14 litres c) 16 litres d) 20 litres

11) The amount of wheat @ 610 per quintal which should be added to 126 quintals of wheat costing Rs. 285 per quintal so that 20% may be gained by selling the mixture at Rs. 480 per quintal will be:

- a) 38 quintals b) 49 quintals c) 69 kg d) 50 kg

12) Three vessels contain equal mixtures of milk & where in the ratio 6 : 15 : 2 & 3 : 1 respectively. If all the solutions are mixed together, the ratio of milk to water in the final mixture will be:

- a) 64 : 65 b) 65 : 64 c) 19 : 65 d) 65 : 19

13) Three vessels contain equal mixtures of milk & where in the ratio 6 : 15 : 2 & 3 : 1 respectively. If all the solutions are mixed together, the ratio of milk to water in the final mixture will be:

- a) 64 : 65 b) 65 : 64 c) 19 : 65 d) 65 : 19

14) Three vessels contain equal mixtures of milk & where in the ratio 6 : 15 : 2 & 3 : 1 respectively. If all the solutions are mixed together, the ratio of milk to water in the final mixture will be:

- a) 64 : 65 b) 65 : 64 c) 19 : 65 d) 65 : 19

ratio of milk and water when he has finished delivering at the third house?

- a) 64 : 61 b) 27 : 37 c) 16 : 19 d) 19 : 27

8) Several litres of acid were drawn off from a 54 litres vessel full of acid and an equal amount of water is added. Again the same volume of the mixture was drawn off and replaced by water. As a result, the vessel contained 24 litres of pure acid. How much acid was drawn off initially ?

- a) 12 litres b) 16 litres c) 18 litres d) 24 litres

9) A mixture of a certain quantity of milk with 16 litres of water is worth 90 paise per litre. If pure milk be worth 108 paise per litre, how much milk in there is the mixture ?

- a) 60 litres b) 20 litres c) 80 litres d) 90 litres

10) A mixture of a certain quantity of milk with 32 litres of water is worth Rs. 1.50 per litre. If pure milk be worth Rs. 4.50 per litre, how much milk is there in the mixture?

- a) 18 litres b) 14 litres c) 16 litres d) 20 litres

11) The amount of wheat @ 610 per quintal which should be added to 126 quintals of wheat costing Rs. 285 per quintal so that 20% may be gained by selling the mixture at Rs. 480 per quintal will be:

- a) 38 quintals b) 49 quintals c) 69 kg d) 50 kg

12) Three vessels contain equal mixtures of milk & where in the ratio 6 : 15 : 2 & 3 : 1 respectively. If all the solutions are mixed together, the ratio of milk to water in the final mixture will be:

- a) 64 : 65 b) 65 : 64 c) 19 : 65 d) 65 : 19

13) To mix spirits worth Rs. 8, Rs. 6 and Rs. 3 per gallon for making a mixture worth Rs. 5 per gallon, how much of each quantity is to be taken ?

- a) 1 : 1 : 2 b) 2 : 1 : 2 c) 1 : 2 : 1 d) 1 : 2 : 2

14) Two equal containers are filled with a mixture of water and alcohol. One of them contains three times as much alcohol as the other. The mixtures in the two containers are then mixed and it is found that the ratio

a) $2 : 1, 3 : 4$ b) $1 : 3, 1 : 2$ c) $2 : 3, 4 : 3$ d) None

15) Two equal containers are filled with a mixture of water and alcohol. One of them contains three times as much alcohol as the other. The mixtures in the two containers are then mixed and it is found that the ratio

a) 16 litres b) 18 litres c) 16.8 litres d) 16.2 litres

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) B | 2) C | 3) D | 4) A |
| 5) D | 6) D | 7) A | 8) C |
| 9) C | 10) C | 11) C | 12) D |
| 13) A | 14) D | 15) D | |

Simple and Compound Interest

Simple Interest

If P stands for principal, R the rate percent per annum, T the number of years. A the amount and St the simple interest, then

$$\begin{aligned} \text{(i)} \quad SI &= \frac{PRT}{100} & \text{(ii)} \quad R &= \frac{SI \times 100}{P \times T} & \text{(iii)} \quad T &= \frac{SI \times 100}{P \times R} \\ \text{(iv)} \quad P &= \frac{SI \times 100}{R \times T} & \text{(v)} \quad SI &= \frac{ART}{100 + RT} & \text{(vi)} \quad P &= \frac{100 \cdot A}{100 + RT} \end{aligned}$$

If a principal P becomes 'a' times of itself in T years at R% rate per annum, then $RT = (a - 1) \Psi 100$

If a principal P become 'd' times of itself in T_1 years at $R_1\%$ rate per annum and 'b' times in T_2 years at $R_2\%$, then

$$\text{(i)} \quad \frac{a - 1}{T_1} = \frac{b - 1}{T_2} \quad \text{(ii)} \quad \frac{a - 1}{R_1} = \frac{b - 1}{R_2}$$

If a person deposits Rs. x_1 in a bank at $r_1\%$ per annum and Rs. x_2 at per annum, then the interest for the whole sum is

$$\left[\frac{x_1 r_1 + x_2 r_2}{x_1 + x_2} \right] \%$$

If a certain principal amounts to Rs. A_1 in t_1 years and to Rs. A_2 in t_2 years, then the sum is given by Rs. $\left(\frac{A_2 t_1 - A_1 t_2}{t_1 - t_2} \right)$ and the rate%

per annum is given by $\left[\frac{100(A_2 t_1 - A_1 t_2)}{(A_1 t_2 - A_2 t_1)} \right] \%$

If two equal principals are deposited for t_1 and t_2 years at $r_1\%$ and $r_2\%$ per annum such that the difference between their interests is D, then the principal can be obtained by

$$P = \text{Rs.} \frac{D \times 100}{r_1 t_1 - r_2 t_2}$$

Ex. What annual installment will discharge a debt of Rs. 2150 due in 4 year at 5% per annum at simple interest?

Sol. Let the first installment be = Rs. 100

then the second installment = Rs. $(100 + 5) = \text{Rs. } 105$

the third installment = Rs. $(105 + 5) = \text{Rs. } 110$

the fourth installment = Rs. $(110 + 5) = \text{Rs. } 115$

$\therefore$ the total debt to be paid = $100 + 105 + 110 + 115 = \text{Rs. } 430$

If the debt is Rs. 430 then first installment = Rs. 100

If the debt is Rs. 2150 then first installment =

$$= \text{Rs. } \frac{100 \times 2150}{430} = \text{Rs. } 500$$

Compound Interest

If P be the principal, A be the amount, R the rate % per annum, T the number of years and CI the compound interest then

$$(i) A = P \left(1 + \frac{R}{100} \right)^T$$

$$(i) CI = A - P = P \left[\left(1 + \frac{R}{100} \right)^T - 1 \right]$$

If the interest is compound half-yearly the rate % is halved and time period is doubled i.e.,

$$A = P \left(1 + \frac{R/2}{100} \right)^{2T}$$

If the difference between the SI and CI is Rs. D on the principal P at R% rate per annum then

$$\text{For 2 years} \Rightarrow D = P \left(\frac{R}{100} \right)^2$$

$$\text{For 3} \Rightarrow \text{years} = D = P \left(\frac{R}{100} \right)^2 \left(\frac{300 + R}{100} \right)^2$$

If the principal becomes 'n' times in 't' years then the rate% per annum of compound interest is given by

$$R = 100 [(n)^{1/t} - 1]\%$$

If the SI on the principal P at R% rate per annum for T years is given, then the CI on the same conditions is given by

$$CI = SI \left[1 + \frac{R}{100} \right]$$

Sample Questions

1) A sum doubles in 20 years at simple interest. How much is the rate ?

- a) 5% p.a. b) 4% p.a. c) 8% p.a. d) 10% p.a.

2) A sum of Rs. 5984 becomes Rs. 8976 in 6 years at SI. What is the rate ?

- a) 10% p.a. b) 6% p.a. c) 12% p.a. d) 8.33% p.a.

3) The simple interest on a certain principal @ 4% p.a. for 5 years is Rs. 800. How much is the principal 3. amount?

- a) Rs. 5000 b) Rs. 6000 c) Rs. 4500 d) Rs. 4000

4) A certain sum of money becomes Rs. 1250 in a span of 5 years at simple interest and further to Rs. 1700 in the span of 8 years. At the same rate, what would it amount to at the end of 12 years?

- a) Rs. 2250 b) Rs. 2050 c) Rs. 2300 d) Rs. 2450

5) The simple interest earned on a certain sum of money for 10 years at the rate of 5% p.a. was half the sum. How much is the sum ?

- a) Rs. 1000 b) Rs. 2000 c) Rs. 3000 d) Rs. 4000

6) An amount of Rs. 8600 is to be divided in two parts such that the simple interest on the first part at 15% p.a. for 4 years is equal to the simple interest on the second part at 20% p.a. for 3 years. What is the difference between the two parts?

- a) Rs. 500 b) Rs. 400 c) Rs. 200 d) Nil

7) Rs. 800 becomes Rs. 956 in 3 years at certain simple rate of interest. If the rate of interest is increased by 4% what amount will Rs. 800 become in 3 years ?

- a) Rs. 1020.8 b) Rs. 1025 c) Rs. 1092 d) None

8) Srinivasan invests two equal amounts in two banks giving 10% and 12% rate of interest respectively. At the end of year the interest earned is Rs. 1650. Find the sum invested in each.

- a) Rs. 8,500 b) Rs. 15,000 c) Rs. 7,500 d) Rs. 17,000

9) A certain amount of money is invested at the simple interest of 15% pe annum. If it had been invested at compound interest, an extra interest of Rs. 450 would have been obtained in the second year. What must be the amount invested?

a) Rs. 30,000 b) Rs. 15,000 c) Rs. 20,000 d) Rs. 10,000

10) What is the difference between the CI and SI on a sum of Rs. 1600 at 5% p.a. for period of 2 years?

a) Rs. 8 b) Rs. 4 c) Rs. 16 d) Rs. 12

11) What is the difference between the compound interest and the simple interest on a capital of Rs. 16,000 at the rate of 15% per annum for a period of 2 years?

a) Rs. 420 b) Rs. 720 c) Rs. 180 d) Rs. 360

12) At what rate of interest per annum would the difference between the compound interest and the simple interest at the end of two years on the capital of Rs. 60,000 be Rs. 1944?

a) 22% b) 20% c) 18% d) 16%

13) The income of a company increases 20% per annum. If its income is Rs. 26,64,000 in the year 1999, what was its income in the year 1997?

a) Rs. 28,20,000 b) Rs. 18,50,000
c) Rs. 21,21,000 d) Rs. 28,55,000

14) If the difference between compound interest, compounded every six months, and the simple interest on a certain sum of money at the rate of 12% per annum for one year is Rs. 36, the sum is

a) Rs. 10,000 b) Rs. 12,000 c) Rs. 15,000 d) Rs. 9,000

15) Balan borrowed Rs. 1,000 at 10 percent per annum simple interest. He immediately lent the whole sum at 10 percent per annum compound interest. At the end of 2 years, he would gain

a) Rs. 1 b) Rs. 10 c) Rs. 11 d) Rs. 100

Solutions

1) A 2) D 3) D 4) C

5) C

6) D

7) D

8) C

9) C

10) B

11) D

12) C

13) B

14) A

15) B

Time And Distance

If S km/hr be the speed, D km be the distance and T hour be the time, then

$$(i) \quad S = \frac{D}{T} \qquad (ii) \quad T = \frac{D}{S} \qquad (iii) \quad D = S \times T$$

If a person travels a distance at S_1 km/h and returns S_2 km/h, the the average speed is given by

$$\text{Average Speed} = \frac{2S_1 \cdot S_2}{S_1 + S_2} \text{ km/h.}$$

If a person changes his speed to $\left(\frac{x}{y}\right)$ of usual speed and reaches of usual speed and reaches early by z min, then the usual time taken

by him is given by, usual time = $\frac{z}{1 - \frac{y}{x}}$ and in case of late by z

min it is $\frac{z}{\frac{y}{x}}$

If two persons start from a place walking at S_1 km/h and S_2 km/h respectively then at the end of t hours.

$(y - x)t$ km a part of they more in same direction

$(x + y)t$ km a part of they more in opposition direction

Time taken by a train of length L_1 m moving at a speed of S km/h to cross a bridge/tunnel/platform of length L_2 m is given by

$$T = \frac{L}{S} \times \left(\frac{18}{5}\right) \text{ sec.}$$

Time taken by a train of length L_1 m moving at a speed of S km/h to cross a bridge/tunnel/platform of length L_2 m is given by

$$T = \frac{L_1 + L_2}{S} \times \left(\frac{18}{5}\right) \text{sec.}$$

$$T = \frac{L_1}{S_1 + S_2} \times \left(\frac{18}{5}\right) \text{sec.}$$

Time taken by a train of length L_1 m moving at a speed of S_1 km/h to cross another train of length L_2 m moving at a speed of S_2 km/h in opposite directions is given by

$$T = \frac{L_1 + L_2}{S_1 + S_2} \times \left(\frac{18}{5}\right) \text{sec.}$$

If the train is late in first case and reaches early in second case vice versa take (+) sign

If the speed of the boat is S_1 km/h and that of the stream/current is S_2 km/h then

$$\text{Upstream Speed} = (S_1 - S_2) \text{km/h}$$

$$\text{Upstream Speed} = (S_1 + S_2) \text{km/h}$$

If the upstream speed is S_1 km/h and downstream speed is S_2 km/h, then

$$\text{Speed of the boat} = \frac{S_1 + S_2}{2} \text{ km/h}$$

$$\text{Speed of the stream/current} = \frac{S_1 - S_2}{2} \text{ km/h}$$

Sample Questions

1) Anuva takes 20 minutes less to reach her office if her speed increase by 5 km/h and takes 30 minutes more if her speed decreases by 5 km/h. What is her original speed?

- a) 25 km/h b) 30 km/h c) 35 km/h d) None

2) A boat takes 20 minutes and 30 minutes to cover a particular distance downstream and upstream respectively. If the speed of the boat in still water is 20 m/s, find the speed of the stream?

- a) 6 m/s b) 8 m/s c) 12 m/s d) 4 m/s

3) 100 m long train crosses a stationary man in 10 seconds. What is the speed of the train?

- a) 18 km/h b) 27 km/h c) 36 km/h d) 45 km/h

4) The speed of boat in still water is 12 km/h. If it takes 1 hour to go upstream 6 kms; in what time will it return same distance downstream ?

- a) 20 mins b) 30 mins c) 5 mins d) 10 mins

5) Two trains A and B, travelling in opposite directions, cross each other in six seconds. The speed of train A is 126 km/h, while that of train B is 90 km/h. If the length of train A is 160 metres, what is the length of train B?

- a) 100 m b) 80 m c) 180 m d) 200 m

6) A train crosses a stationary person in 3 seconds while it takes 10 seconds to cross a platform, travelling at same speed. If the length of the platform is 210 metres, what is the length of the train?

- a) 180 m b) 135 m c) 90 m d) 45 m

7) A 225 m long train crosses another 150 m long train in 12.5 sec. If one of the trains travels @72 km/h, what is the speed of the other?

- a) 54 km/h b) 27 km/h c) 36 km/h d) 45 km/h

8) What is the speed of a train which overtakes a man walking at a speed of 5 km/h in 30 seconds, if the train is 274 metres long?

- a) 51.88 km/h b) 47.88 km/h c) 37.88 km/h d) 21.67 km/h

9) Two persons, 27 km apart, setting out at the same time are together in 9 hrs if they walk in the same direction, but in 3 hrs. if they walk in opposite directions. What are their rates of walking (in km/h)?

- a) 5, 2 b) 4, 3 c) 6, 3 d) 8, 7

10) A train starts from station X at the rate of 60 km/h and reaches station Y in 45 minutes. If the speed is reduced by 6 km/h.

How much more time will the train take to return from station Y to station X ?

- a) 5 mins b) 7.5 mins c) 35 km/h d) None

11) In one hour a boat goes 11 km along the stream and 5 km against the stream. The speed of the boat in still water (in is: km/h)

- a) 5 b) 6 c) 8 d) 9

12) Find the speed of train in km/h whose length is 200 m and crosses a platform of length 240 m in 22 seconds?

- a) 60 b) 48 c) 53 d) None

13) A train running at speed of 120 km/h crosses a signal post in 15 seconds. What is the length of the train in metres ?

- a) 300 b) 200 c) 500 d) 400

14) A boat has to run upstream to reach point Y from a point X, 20 kms away. The boat starts from point X for point Y and comes back to point X in a total time of 41 minutes and 40 seconds, What is the speed of the boat ?

- a) 166 km/h b) 7.2 km/h
c) 148 km/h d) Data inadequate

15) A and B are two stations 500 km apart. A train starts from A and moves towards B at the rate of 20 km/h. Another train starts from B at the same time and moves towards A at the rate of 30 km/h. How far from A will they cross each other?

- a) 40 km b) 200 km c) 300 km d) 120 km

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) D | 2) D | 3) C | 4) A |
| 5) D | 6) C | 7) C | 8) C |
| 9) C | 10) A | 11) C | 12) D |
| 13) C | 14) D | 15) B | |

Time And Work

$$\frac{M_1 - D_1 - H_1}{W_1 W a_2} = \frac{M_2 - D_2 - H_2}{W_2 W a_1}$$

where M_1, M_2 are the number of men, D_1, D_2 are the number of days, H_1, H_2 are the number of hours, W_1, W_2 are the works and $W a_1, W a_2$ are the wages.

If M_1 men had provisions for P_1 days and after P_2 and M_2 men joined or left, then the provision now lasts for D days is given as

$$D = \frac{M_1 (p_1 - p_2)}{M_1 \pm M_2} \text{ (+ if men joined, - if men left)}$$

If x_1 men or y_1 women can reap a field in D days, then x_2 men and y_2 women would reap it in $\frac{D(x_1 - y_1)}{x_2 y_1 + x_1 y_2}$

If A can do a piece of work in n days then the piece of work completed by A in one day is $\frac{1}{n}$.

If A completes a piece of work in n days and B in m days then they both working together can complete the work in

$$\left(\frac{nm}{n + m} \right) \text{ days.}$$

If A completes a piece of work in n days, B in m days and C in p days then they all working together can complete it in

$$\left(\frac{nmp}{nm + mp + pn} \right) \text{ days.}$$

If A and B complete a piece of work in n days, B and C in m days and C and A in p days, then they all working together can

$$\text{complete it in } \left(\frac{2mnp}{mn + np + mp} \right) \text{ days.}$$

If A is n times good as workman as B, such that together they finish the work in x days, then A and B separately can do the same work in $\left(\frac{n+1}{n}\right)x$ days and $(n+1)x$ days respectively.

Two pipes can fill an empty reservoir in t_1 and t_2 min respectively. If both the pipes are opened simultaneously then the time after which the second pipe is closed so that the total time taken to fill the reservoir is T min, is given by $\left(1 + \frac{T}{t_2}\right)t_2$ min.

There is a hole in a reservoir which empties it in T_1 hours. If a tap is turned on which admits the water in the reservoir at the rate of x litres / hour due to which the reservoir is now emptied in T_2 hours, then the volume of the reservoir is given by

$$\left[\frac{x(T_1 T_2)}{(T_2 - T_1)} \right] \text{ litres.}$$

Sample Questions

1) Ravi and Sonu can do a job alone in 20 days and 30 days respectively. In how many days the job will be finished, if they work together?

- a) 10 b) 12 c) 8 d) 13

2) A and B can do a work in 72 days. B and C in 120 days. A and C in 90 days. In how many days can A alone do the work?

- a) 80 b) 100 c) 110 d) 120

3) If 40 men can complete a work in 10 days working 10 hours a day, then find the number of men is required the work in 5 days if they work 10 hours in a day.

- a) 100 b) 60 c) 150 d) 80

4) A and B undertake to do a piece of work for Rs. 600. A alone can do it in 6 days while B alone can do it in 8 days but with the help of C, they finish it in 3 days. Find the share of C.

- a) Rs. 75 b) Rs. 90 c) Rs. 110 d) Rs. 85

5) 45 men can complete a work in 16 days. Six days after they started working, 30 more men joined them. How many days will they now take to complete the remaining work?

- a) 5 b) 6 c) 7 d) 8

6) X can do $\frac{1}{4}$ of a work in 10 days. Y Can do 40% of the work in 40 days and Z can do $\frac{1}{3}$ of the work in 13 days. Who will complete the work first?

- a) X b) Y c) Z d) X and Z both

7) A can do a certain work in the same time in which B and C together can do it. If A and B together could do it in 10 days and C alone in 50 days, then B alone could do it in

- a) 15 b) 20 c) 25 d) 30

8) Two pipes A and B can fill a tank in 24 min. and 32 minutes respectively. If both the pipes are opened together, after how much time B should be closed so that the tank is full in 18 minutes.

- a) 7 mins b) 6 mins c) 8 mins d) 9 mins

9) Two pipes A and B can fill a tank in 2 hours and 75 minutes respectively. There is also an outlet C. If all the three pipes are opened together then the tank is full in 50 minutes. How much time will be taken by C alone to empty the full tank?

- a) 90 mins b) 10 hours c) 150 mins d) 120 mins

10) Two pipes can fill a cistern in 14 hours and 16 hours respectively. The pipes are opened simultaneously and it is found that due to leakage in the bottom it took 32 min. more to fill the cistern. When the cistern is full, in what time will the leak empty it?

- a) 100 hours b) 111 hours c) 112 hours d) 114 hours

11) Twenty women can do work in sixteen days. Sixteen men can complete the same work in fifteen days. What is the ratio between the capacity of a man and a woman?

- a) 3 : 4 b) 4 : 3 c) 5 : 3 d) 3 : 5

12) A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every 3rd day?

- a) 12 b) 15 c) 16 d) 18

13) A pipe can fill a tank in 15 hours. Due to leak in the bottom, it is filled in 20 hours. If the tank is full, how much time will the leak take to empty it?

- a) 120 hours b) 80 hours c) 40 hours d) 60 hours

14) In 4 men reap 40 acres in 30 days. How many acres will 18 men reap in 12 days?

- a) 70 b) 72 c) 60 d) 68

15) A can do a certain job in 12 days. B is 60% more efficient than A. How many days does B alone take to do the same job?

- a) $91/2$ days b) $81/2$ days c) $71/2$ days d) $111/2$ days

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) B | 2) D | 3) D | 4) A |
| 5) B | 6) C | 7) C | 8) C |
| 9) B | 10) C | 11) B | 12) B |
| 13) D | 14) B | 15) C | |

Permutations & Combinations

Factorial Notation: Let n be a positive integer. Then, factorial n , denoted $n!$ is defined as: $n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1$.

Examples: We define $0! = 1$.

$$4! = (4 \times 3 \times 2 \times 1) = 24.$$

$$5! = (5 \times 4 \times 3 \times 2 \times 1) = 120.$$

Permutations:

The different arrangements of a given number of things by taking some or all at a time, are called permutations.

Examples:

All permutations (or arrangements) made with the letters a, b, c by taking two at a time are (ab, ba, ac, ca, bc, cb) .

All permutations made with the letters a, b, c taking all at a time are: $(abc, acb, bac, bca, cab, cba)$

Number of Permutations:

Number of all permutations of n things, taken r at a time, is given

$$\text{by: } {}^n p_r = n(n-1)(n-2) \dots (n-r+1) = \frac{n!}{(n-r)!}$$

Examples:

$${}^6 p_2 = (6 \times 5) = 30.$$

$${}^7 p_3 = (7 \times 6 \times 5) = 210.$$

Cor. number of all permutations of n things, taken all at a time $= n!$

An Important Result:

If there are n subjects of which p_1 are alike of one kind; p_2 are alike of another kind; p_3 are alike of third kind and so on and p_r are alike of r^{th} kind, such that $(p_1 + p_2 + \dots p_r) = n$.

Then, number of permutations of these n objects is =

$$\frac{n!}{(p_1!)(p_2!)(\dots)(p_r!)}$$

Combinations: Each of the different groups or selections which can be formed by taking some or all of a number of objects is called a combination.

Examples:

Suppose we want to select two out of three boys A, B, C. Then, possible selections are AB, BC and CA.

Note: AB and BA represent the same selection.

All the combinations formed by a, b, c taking ab, bc, ca.

The only combination that can be formed of three letters a, b, c taken all at a time is abc.

Various groups of 2 out of four persons A, B, C, D are:

AB, AC, AD, BC, BD, CD.

Note: that ab ba are two different permutations but they represent the same combination.

Number of Combinations:

The number of all combinations of n things, taken r at a time is:

$${}^nC_r = \frac{n!}{(r!)(n-r)!} = \frac{n(n-1)(n-2) \dots \text{to } r \text{ factors}}{r!}$$

Note:

$${}^nC_n = 1 \text{ and } {}^nC_0 = 1$$

$${}^nC_n = {}^nC_{(n-r)}$$

Examples:

$${}^{11}C_4 = \frac{(11 \times 10 \times 9 \times 8)}{(4 \times 3 \times 2 \times 1)} = 330$$

$${}^{16}C_{13} = {}^{16}C_{(16-13)} = {}^{16}C_3 = \frac{16 \times 15 \times 14}{3!} = \frac{16 \times 15 \times 14}{3 \times 2 \times 1} = 560$$

Type 1

When letters are repeated:

Step 1: Arrange in factorial notation

Step 2: Divide by the factorial of the number of times each letter is repeated

Example: Find the number of ways the letters in the word 'LEADER' can be arranged:

Step 1: $6!$ (since there are 6 letters in total)

Step 2: $6! / 2!$ (since 'E' is repeated twice)

Type 2

When it's about grouping

For Example: Find the number of ways the letters of the word 'LEADING' can be arranged such that the vowels always come together.

Step 1: Group the vowels EAI as a single "letter." So, EAI can be arranged in $3!$ ways.

Step 2: Now consider the remaining letters LDNG as four distinct letters. Together with EAI as one unit, we have 5 "letters" to arrange: LDNG (EAI).

Total ways = $5! \times 3!$

Sample Questions

1) There are 4 different types of chocolates and 5 different types of ice creams, in how many ways can a person buy a chocolate and an ice cream ?

- a) 9 b) 20 c) 15 d) 45

2) How many ways can the letters of the word "ORANGE" be arranged ?

- a) 120 b) 240 c) 60 d) 720

3) How many ways can the letters of the word "MISSISSIPPI" be arranged ?

- a) $11!$ b) $11! / (4! \times 4! \times 2!)$
c) $10! / 8!$ d) $10! / (4! \times 4!)$

4) How many ways can the letters of the word "ORANGE" be arranged , in such a way that O and A are always together ?

- a) 120 b) 240 c) 360 d) 480

5) What is the rank of a word TIGER, if all possible permutations of the word TIGER are arranged in dictionary order.

- a) 99 b) 111 c) 110 d) 109
- 6) In how many ways can 6 people be seated at a round table?
a) 99 b) 120 c) 116 d) 109
- 7) Find the number of ways in which 4 stones, chosen from 7 different stones can be placed on a necklace .
a) 210 b) 1050 c) 105 d) 420
- 8) 8 directors, the vice chairman and the chairman are to be seated around a circular table. If the chairman should sit between a director and the vice chairman, in how many ways can they be seated?
a) $8!$ b) $7! \times 2$ c) $9! \times 2$ d) $8! \times 2$
- 9) A four digit number is formed using digits 1, 3, 5, and 7 without repeating any one of them. What is the sum of all such possible numbers?
a) 106656 b) 666660 c) 106666 d) 105556
- 10) In how many ways 8 ice creams can be selected from 15 non-identical ice creams?
a) $8 C 15$ b) $15 C 7$ c) $14!$ d) 1
- 11) In a group of 6 boys and 4 girls, four children are to be selected. In how many different ways can they be selected such that at least one boy should be there?
a) 159 b) 209 c) 201 d) 212
- 12) How many 3 digit number can be formed with the digits 5, 6, 2, 3, 7 and 9 which are divisible by 5 and none of its digit is repeated?
a) 12 b) 16 c) 20 d) 18
- 13) There are 4 bananas, 7 apples and 6 mangoes in a fruit basket. In how many ways can a person make a selection of fruits from the basket. (Assume all the fruits are similar)
a) 269 b) 280 c) 279 d) 256
- 14) There are 15 points in a plane out of which 6 are collinear. Find the number of lines that can be formed from 15 points.

an ice cream ?

- a) 105 b) 90 c) 91 d) 95

15) Find the total number of distinct vehicle numbers that can be formed using two letters followed by two numbers. Letters need to be different.

- a) 4500 ways b) 54,000 ways c) 64,000 ways d) 65,000 ways

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) B | 2) D | 3) B | 4) B |
| 5) B | 6) B | 7) C | 8) D |
| 9) A | 10) B | 11) B | 12) C |
| 13) C | 14) C | 15) C | |

Probability

• **Random Experiment:** Probability is the study of random or non deterministic experiments. If the dice is tossed in the air, then it is certain that the dice will come down, but is not certain that, say a 3 will appear.

Definition: A random experiment is an experiment whose result would not be predicted but the list of possible outcomes are known. The non predicted outcomes cannot be taken under random experiments. The result of random experiments may not be predicted exactly but the result must be with in the list of predicted outputs.

Example: 1) Tossing a fair coin.

2) Rolling a dice is a random experiment, since its results could not be predicted in any trial.

3) Selection of a plastic component and verification of its compliance.

4) Life time of a computer.

5) Number of calls to a communication system during a fixed length interval of time.

• **Outcome:** The result of a random experiment will be called an outcome.

Example: 1) Tossing a coin. The result is either Head(H) or Tail(T).

2) In an experiment of throwing a six-faced dice. The possible outcomes are 1, 2, 3, 4, 5 and 6.

• **Sample Space:** The set of all possible outcomes of some given experiment is called sample space. A particular outcome, i.e. an element in that set is called a sample point or sample.

Example: 1) Toss a dice and observe the number that appears on top. Then the sample space consists of the six possible numbers: $S = \{1, 2, 3, 4, 5, 6\}$

2) Toss a coin 2 times and observe the sequence of heads (H) and tails (T) that appears.

Then the sample space S consists of four elements: $S = \{HH, HT, TH, TT\}$

Note: Shortcut: Tossing a coin 2 times is same as tossing 2 coins at a time.

$S = \{H, T\} \times \{H, T\}$ $S = \{HH, HT, TH, TT\}$ 3) Toss a coin until a head appears and then count the number of times the coin was tossed. The sample space of this experiment is $S = \{1, 2, 3, \dots, \infty\}$. Here ∞ refers to the case when a head never appears and so the coin is tossed an infinite number of times. This is an example of a sample space which is countably infinite.

• **Events:** An event A is a set of outcomes or, in other words, a subset of the sample space S .

Example: If A random experiment is associated with what is the day today. It may be from Sunday to Saturday. If today is Friday and Friday belongs to the sample space $S = \{\text{Sun, Mon, Tue, Wed, Thu, Fri, Sat}\}$.

Different Types of Events

• **Simple or Elementary Events:** An event with only one sample point is called simple or elementary event. In an experiment of tossing three coins at a time, the event 'A' is that all coins turn up with heads consists of only one point HHH. Then 'A' is a simple event. As a matter of fact each outcome of an experiment is a simple event.

• **Complimentary Event:** An event $\bar{A}$ (or A_1) is said to be complementary to an event 'A' in sample space 'S' consists of all those points which are not in 'A'.

Example: In tossing a coin three times, sample space S consists of eight points.

$S = \{HHH, HHT, HTH, THH, HTT, TTH, THT, TTT\}$ The event 'A' is such that there should be no heads in the sample point is $\{TTT\}$.

Then the event $\bar{A}$ (or A_1) complementary to 'A' is that there exists at least one head in the sample space i.e. (HHH), (HHT), (HTH), (THH), (HTT), (TTH), (THT).

• **Equal Events:** Two events A and B are said to be equal if $A \subseteq B$ and $B \subseteq A$. This statement implies that all the points of A are also the points of B and vice-versa.

Example: Let sample space $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Let A be the event defined as 'even number' and event B is defined as 'multiplies of 2'.

Then $A = \{2, 4, 6, 8, 10\}$; $B = \{2, 4, 6, 8, 10\}$ here every point in A is also a point in B and vice-versa. Therefore events A and B are said to be equal events.

• **Transitivity of events:** If A, B and C are 3 events such that $A \subseteq B$ and $B \subseteq C$ it implies that $A \subseteq C$ such a property of events is known as transitivity of events.

Let the sample space $S = \{1, 2, \dots, 100\}$.

Event A be the 'even numbers': $A = \{2, 4, 6, 8, \dots, 100\}$

Event B be the 'multiples of 4': $B = \{4, 8, 12, \dots, 100\}$

Event C be the 'multiples of 8': $C = \{8, 16, 24, \dots, 100\}$

Event point in C is also point in B and event point in B is also point in A but not vice-versa i.e. $A \subseteq B \subseteq C$.

• **Compound event:** An event which is not simple or elementary is called a compound event. Every compound event can be uniquely represented by the union of a set of elementary events.

Sample Questions

1) Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?

- a) $1/2$ b) $2/5$ c) $8/15$ d) $9/20$

2) A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue?

- a) $10/21$ b) $11/21$ c) $2/7$ d) $5/7$

3) In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green?

- a) $1/3$ b) $3/4$ c) $7/19$ d) $8/21$

4) What is the probability of getting a sum 9 from two throws of a dice?

- a) $1/6$ b) $1/8$ c) $1/9$ d) $1/12$

5) Three unbiased coins are tossed. What is the probability of getting at most two heads?

- a) $3/4$ b) $1/4$ c) $3/8$ d) $7/8$

6) Two dice are thrown simultaneously. What is the probability of getting two numbers whose product is even?

- a) $3/4$ b) $1/4$ c) $3/8$ d) $7/8$

7) In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is:

- a) $21/46$ b) $25/117$ c) $1/50$ d) $3/25$

8) In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize?

- a) $1/10$ b) $2/5$ c) $2/7$ d) $5/7$

9) From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings?

- a) $1/15$ b) $25/57$ c) $35/256$ d) $1/221$

10) Two dice are tossed. The probability that the total score is a prime number is:

- a) $1/6$ b) $5/12$ c) $1/2$ d) $7/9$

11) A card is drawn from a pack of 52 cards. The probability of getting a queen of club or a king of heart is:

- a) $1/13$ b) $2/13$ c) $1/26$ d) $1/52$

12) A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red, is:

- a) $1/22$ b) $3/22$ c) $2/91$ d) $2/77$

13) Two cards are drawn together from a pack of 52 cards. The probability that one is a spade and one is a heart, is:

- a) $3 / 20$ b) $29 / 34$ c) $47 / 100$ d) $13 / 102$

14) One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card (Jack, Queen and King only)?

- a) $1 / 13$ b) $3 / 13$ c) $1 / 4$ d) $9 / 52$

15) A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white?

- a) $3 / 4$ b) $4 / 7$ c) $1 / 8$ d) $3 / 7$

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) B | 2) A | 3) A | 4) C |
| 5) B | 6) B | 7) A | 8) C |
| 9) D | 10) B | 11) C | 12) C |
| 13) D | 14) B | 15) B | |

Mensuration

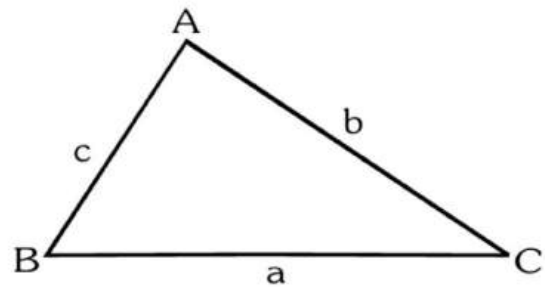
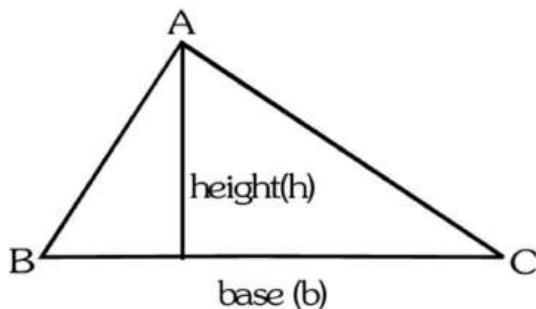
Area and Perimeters

Triangle

$$\text{Area of triangle} = \frac{1}{2} \Psi \text{ base} \Psi \text{ height}$$

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{where, } s = \frac{a+b+c}{2} \text{ (semi-perimeter)}$$



$$\text{Perimeter of triangle} = a + b + c$$

$$\text{Area of an equilateral triangle} = \frac{\sqrt{3}}{4} a^2, \text{ where } a \text{ is its side.}$$

$$\text{Median/altitude/centroid of an equilateral triangle} = \frac{\sqrt{3}}{2} a.$$

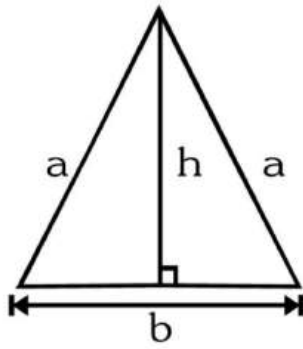
If R be radius of the circumcircle and r be the radius of an incircle of an equilateral triangle then

$$R = \frac{a}{\sqrt{3}}, r = \frac{a}{2\sqrt{3}}, \text{ i.e., } R = 2r$$

$$\text{Perimeter of an equilateral triangle} = 3a$$

$$\text{Area of an isosceles triangle} = \frac{b}{4} \sqrt{4a^2 - b^2}$$

$$\text{Height (h) of an isosceles triangle} = \frac{1}{2} \sqrt{4a^2 - b^2}$$



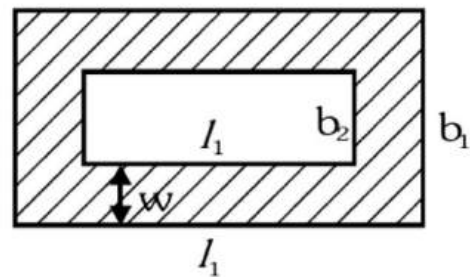
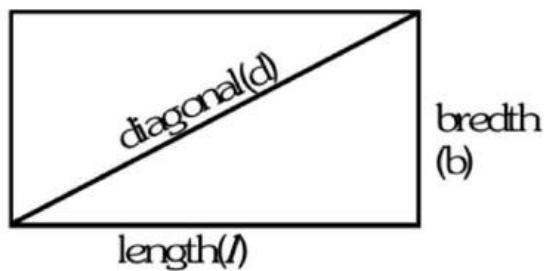
Rectangle

Area of rectangle = length Ψ breadth

Perimeter of rectangle = $2(l + b)$

Length of the diagonal = $\sqrt{l^2 + b^2}$

Area of track = $(l_1 b_1 - l_2 b_2)$

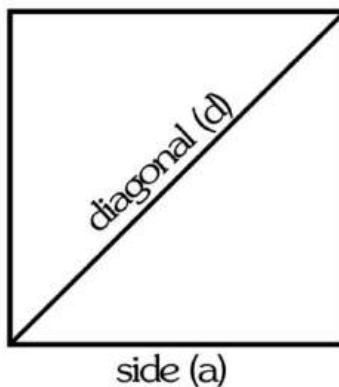


Square

Area of a square = $(\text{side})^2$

Perimeter of a square = $4a$

Diagonal of a square = $\sqrt{2}a$



Circle

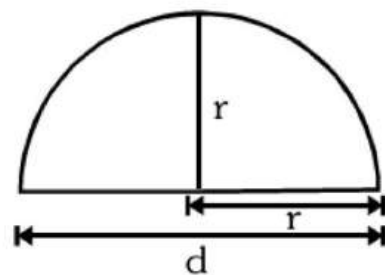
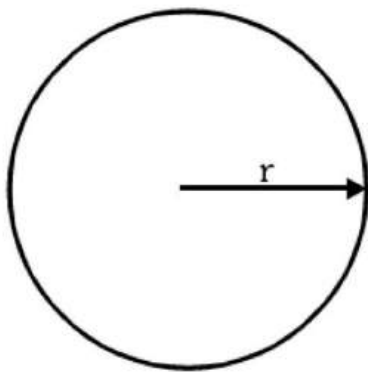
Area of a circle = πr^2

Circumference of a circle = $2\pi r$

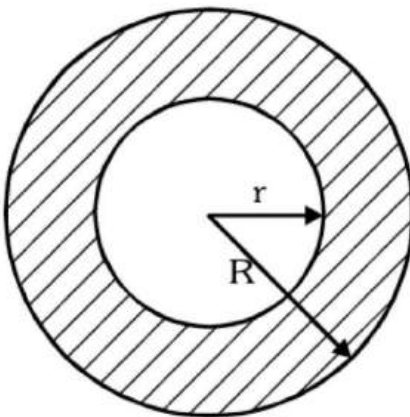
Diameter = 2 Ψ radius

Area of a semicircle = $\frac{1}{2} \pi r^2$

Perimeter of a semicircle = $\pi r + d$



Area of circular path/track/ring = $\pi[R^2 - r^2]$

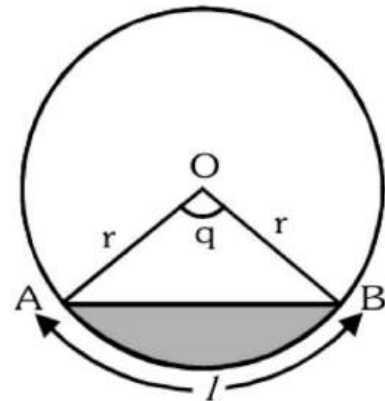
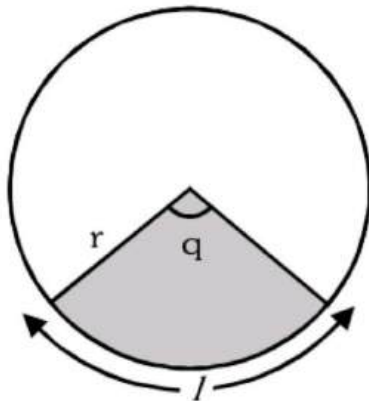


Area of sector of a circle = $\frac{\theta}{360^\circ} \Psi \pi r^2 \frac{1}{2} l r$

Length of the arc (l) = $\frac{\theta}{360^\circ} \Psi 2\pi r$

Area of a segment = Area of sector - Area of $\triangle AOB$

$$\text{Area of } \triangle AOB = \frac{1}{2} r^2 \sin \theta$$



Short cuts:

If the length of a square/rectangle is increased by $x\%$ and the breadth is increased by $y\%$, the net effect on the area is given by

$$\text{Net Effect} = \left[x + y + \frac{xy}{100} \right] \%$$

If the length of a square/rectangle is increased by $x\%$ and the breadth is decreased by $y\%$, the net effect on the area is given by

$$\text{Net Effect} = \left[x - y - \frac{xy}{100} \right] \%$$

If the length and breadth of a square/rectangle is decreased by $x\%$ and $y\%$ respectively, the net effect on the area is given by

$$\text{Net Effect} = \left[-x - y + \frac{xy}{100} \right] \%$$

If the side of a square/rectangle/triangle is doubled the area is increased by 300% i.e., the area becomes four times on itself.

If the radius of a circle is decreased by $x\%$, the net effect on the

area is $\left(-\frac{x^2}{100} \right) \%$, i.e., the area is decreased by $\left(\frac{x^2}{100} \right) \%$.

If a floor of dimensions $l\text{ m} \times b\text{ m}$ is to be covered by a carpet of width $w\text{ m}$, the length of the carpet is $\left(\frac{lb}{w}\right)\text{ m}$.

If a floor of dimensions $l\text{ m} \times b\text{ m}$ is to be covered by a carpet of width $w\text{ m}$ at the rate Rs. X per metre, then the total amount required is Rs. $\left(\frac{Xlb}{w}\right)$.

If a room of dimensions $l\text{ m} \times b\text{ m}$ is to be paved with square tiles then

(i) The side of the square tile = HCF of l and b

(ii) The number of tiles required = $\frac{l \times b}{(\text{HCF of } l \text{ and } b)^2}$

If the sides of a rectangle field of area $X\text{ sq. m}$ are in the ratio $m:n$ then the sides are given by

$$\sqrt{X \cdot \frac{m}{n}} \text{ and } \sqrt{X \cdot \frac{n}{m}}$$

If the side of a regular polygon is a and the polygon has n sides

then the area of the polygon is $\left[\frac{n}{4} \cot\left(\frac{\pi}{n}\right)\right]a^2\text{ sq units}$.

Area of a square inscribed in a circle of radius r is $2r^2$ and the side of a square inscribed in a circle of radius r is $\sqrt{2}r$.

The area of the largest triangle inscribed in a semicircle of radius r is r^2 .

The number of diagonals of a regular polygon of n sides is given by $\frac{n(n-3)}{2}$.

If a square hall x meters long is surrounded by a verandah (on the outside the hall) d meters wide, the area of the verandah is given by $4d(x+d)\text{ sq m}$.

If the verandah is made inside, the area is given by $4d(x-d)\text{ sq m}$.

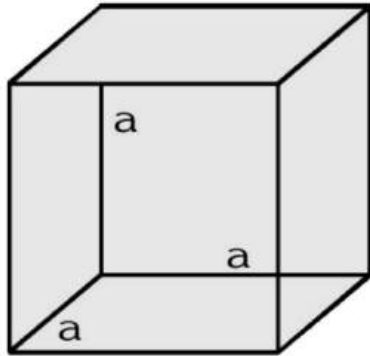
Volume and Surfaces Areas

Cube

Volume of a cube = a^3

Total surface area = $6a^2$

Total surface area = $\sqrt{3}a$



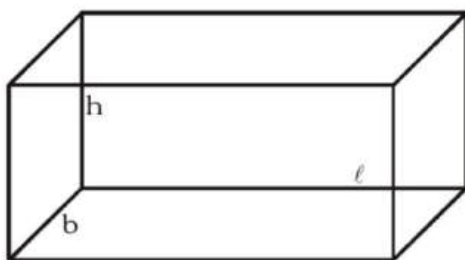
Cuboid

Volume of cuboid = lbh

Total Surface area = $2(lb + bh + lh)$

Length of the diagonal = $\sqrt{l^2 + b^2 + h^2}$

Area of four walls = $2(l + b)h$

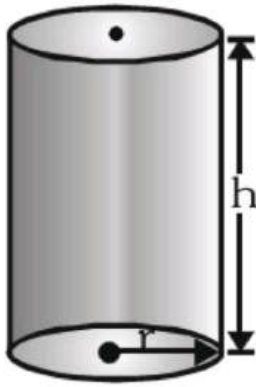


Cylinder

Volume of cylinder = $\pi r^2 h$

Volume of cylinder = $2\pi r(h + r)$

Curved Surface Area = $2\pi r h$



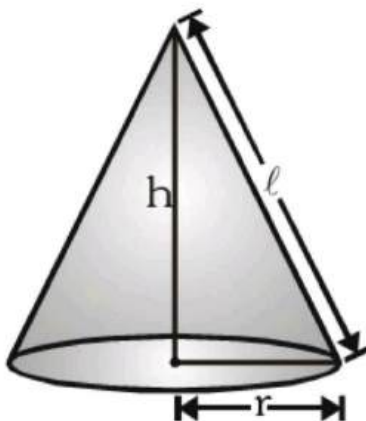
Cone

$$\text{Volume of Cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Total Surface Area} = \pi r(r + l)$$

$$\text{Curved Surface Area} = \pi r l$$

$$\text{Slant Height (l)} = \sqrt{h^2 + r^2}$$



If the side of a cube is increased by x% then its volume increases

$$\text{by } \left[\left(1 + \frac{x}{100} \right)^3 - 1 \right] \times 100\%$$

If length, breadth and height of a cuboid are increased by x%, y% and z% respectively, then its volume is increased by

$$\left[x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{(100)^2} \right] \%$$

Sphere

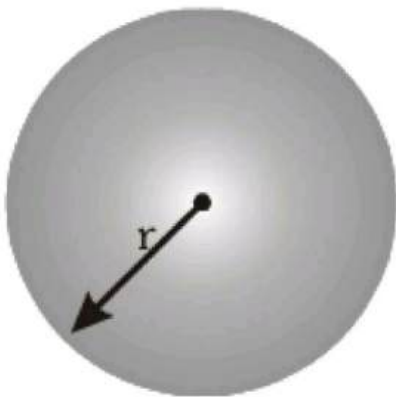
$$\text{Volume of Sphere} = \frac{4}{3}\pi r^3$$

$$\text{Total Surface Area} = 4\pi r^2$$

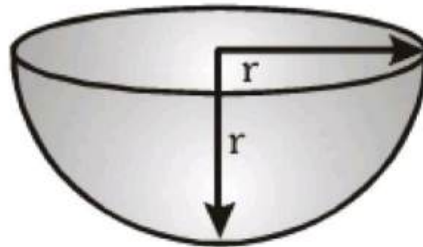
$$\text{Volume of Hemisphere} = \frac{2}{3}\pi r^3$$

$$\text{Total Surface area of Hemisphere} = 3\pi r^2$$

$$\text{Curved Surface area of Hemisphere} = 2\pi r^2$$



Sphere

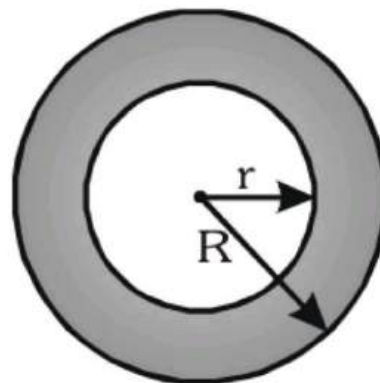
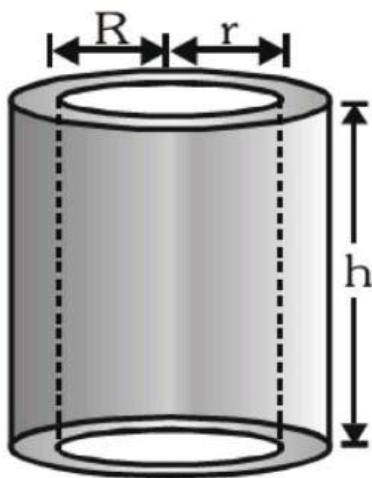


Hemisphere

Hollow Cylinder

$$\text{Volume of Hollow Cylinder} = \pi[R^2 - r^2]h$$

$$\text{Total Outer Surface Area} = 2\pi rh + \pi R^2 + \pi(R^2 + r^2)$$



Hollow Sphere (Shell)

Quick-Aid

Curved Surface Area $\pi[R + r]l$

Total Surface Area $= \pi[(R + r)l + r^2 l + R^2 l]$

Total Surface Area of bucket $= \pi[(R + r)l + r^2 l]$

Curved Surface area of Hemisphere $= 2\pi r^2$

[$\because$ it is open at the bigger end]

If the areas of three adjacent faces of cuboid are A_1 , A_2 and A_3 , the volume of cuboid is given by $\sqrt{A_1 A_2 A_3}$.

If the side of a cube is increased by $x\%$ then its volume increases

by $\left[\left(1 + \frac{x}{100} \right)^3 - 1 \right] \times 100\%$

If length, breadth and height of a cuboid are increased by $x\%$, $y\%$ and $z\%$ respectively, then its volume is increased by

$\left[x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{(100)^2} \right] \%$

If the sides and diagonal of a cuboid are given, then the total surface area in terms of diagonal and sides is given by

Total Surface Area $= (\text{Sum of the sides})^2 - (\text{diagonal})^2$

If the side of a cube is increased by $x\%$, the surface area is

increased by $\left(2x + \frac{x^2}{100} \right) \%$.

If each side of a cube is doubled, its volume becomes 8 times, i.e., volume is increased by 700%.

Sample Questions

1) A wire is in the form of a circle of radius 42 cm. If it is bent into a square, then the side of the square will be:

a) 60 cm

b) 62 cm

c) 64 cm

d) 66 cm

2) The area of a circle is 13.86 hectares. The cost of fencing it at the rate of 60 paise per metre is:

- a) Rs. 784 b) Rs. 788 c) 792 d) 796

3) The sides of a rectangular field are in the ratio 3 : 4 and its area 7500 sq m. The cost of fencing it at 25 paise per metre is

- a) Rs. 77.50 b) Rs. 70.50 c) 67.50 d) None

4) If the perimeter of a square is 16 cm, then the area of the square is

- a) 8 sq. m b) 16 sq. m c) 32 sq. m d) 64 sq. m

5) The perimeter of a square is 12 cm and that of another is 16 cm. The side of a square whose area is equal to the sum of the areas of the above two square would be

- a) 4 cm b) 5 cm c) 5.5 cm d) 6 cm

6) A wire in the form of a circle of radius 98 cm is cut and bent in the form of a square. The side of the square, thus formed is

- a) 146 cm b) 152 cm c) 154 cm d) 156 cm

7) If the radius of a circle increases by 6%, then by how much does the area of the circle increase?

- a) 0.36% b) 3.6% c) 12.36% d) 6%

8) The cost of white washing the four walls of a room is Rs. 25. The cost of white washing a room twice in length, breadth and height each will be

- a) Rs. 50 b) Rs. 75 c) Rs. 120 d) None

9) If length of the rectangle is increased by 50% and breadth is decreased by 20%. Then what is the percentage change in the area?

- a) 20% decrease b) 20% increase
c) 80% increase d) 30% decrease

10) The length of a rectangle is 1 cm more than its width. Its perimeter is 14 cm. The area of the rectangle is

- a) 16 cm² b) 14 cm² c) 12 cm² d) 10 cm²

- 11) If the radius of a circle is doubled, its area is increased by
 a) 100% b) 200% c) 300% d) 400%
- 12) If the ratio of the areas of two square is 16:1, then the ratio of their perimeter is
 a) 4 : 1 b) 16 : 1 c) 1 : 3 d) 3 : 4
- 13) The radius of a wheel is 21 cm. How many revolutions will it make in travelling 924 metres ? (take $\pi = 22 / 7$)
 a) 900 b) 500 c) 200 d) 700
- 14) If the length of a rectangle is increased by 25% and the width is decreased by 20%, then the area of the rectangle
 a) increases by 5% b) decreases by 5%
 c) remains unchanged d) increases by 10%
- 15) If a wire is bent into the shape of a square, the area of the square is 81 sq cm. When the wire is bent into a semicircular shape, the area of the semicircle is
 a) 154 cm² b) 77 cm² c) 44 cm² d) 22 cm²

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) D | 2) C | 3) D | 4) B |
| 5) B | 6) C | 7) C | 8) D |
| 9) B | 10) C | 11) C | 12) A |
| 13) D | 14) C | 15) B | |

Data Interpretation

The information related to any event given in the form of graphs, tables, charts etc is termed as data. The methodology of interpreting data to get the information is known as data interpretation. Mathematical identities which we use in data interpretation are given below. To solve the problems on data interpretation, you need to be thorough in 'Percentages', 'Ratios' and 'Averages' chapters.

Percentage: Proportions with the base 100 are known as percentages (%).

For example, $\frac{x}{y} \times 100$, is in percentage form.

e.g.: If the ratio of boys to total number of students in a college is $\frac{1015}{4060}$. This can be written in a percentage form as $\frac{1015}{4060} \times 100$
 $= 25\%$.

To find by how much percent 'x' is more or less than y (or over y) when compared to y is given as

$$\text{Required Percentage} = \frac{\text{Value of X} - \text{Value of Y}}{\text{Value of Y}} \times 100$$

Observe that the denominator contains the value with which the comparison is made. In the above formula, if numerator is positive, then there is percentage growth. If numerator is negative, then there is a decline in the percentage.

Ratio: In the simplest possible form, ratio is a quotient or the numerical quantity obtained by dividing one figure by the other figure of same units.

Tabular Data Interpretation

Example: Study the following table carefully and answer the questions that follow. Table: Percentage of marks scored by students in SSC

Marks percentage	Girls	Boys
>75	25	12
60-75	15	12
50-59	10	23
35-49	5	2
< 35	3	1

1) Give the total percentage of Girls who wrote SSC examination from that School.

- (A) 25% (B) 54% (C) 23% (D) 58%

2) Give the percentage of students who scored distinction (> 75).

- (A) 43% (B) 34.25% (C) 24.85% (D) 40%

3) Give fail percentage of students in SSC examination.

- (A) 1% (B) 2% (C) 4% (D) 8%

4) Give pass percentage of boys in SSC examination.

- (A) 90% (B) 88% (C) 98% (D) 99%

5) Give the percentage of students who scored more than 60% in the SSC examination.

- (A) 25% (B) 59.3% (C) 22.2% (D) 50%

Explanation: 1) **b**; Total no. of girls appeared for SSC

Examination = 25 + 15 + 10 + 5 + 3 = 58.

Total no. of students appeared for SSC examination = 58 + 50 = 108. $\therefore$ Percentage of girls who wrote SSC Examination =

$$\frac{58}{108} \times 100 = 53.7 = 54\% \text{ (approximately).}$$

2) **b**; No. of students who scored distinction = 25 + 12 = 37. $\therefore$

Percentage of students who scored distinction =

$$\frac{37}{108} \times 100 = 34.25\%.$$

3) **c**; Total no. of students failed in SSC examination = 4.

$$\therefore \text{Fail \%} = \frac{4}{108} \times 100 = 3.7 = 4\% \text{ (approximately).}$$

4) **c**; No. of boys passed in the examination = 49.

$\therefore$ Boys pass percentage =

$$\therefore \text{Fail \%} = \frac{49}{50} \times 100 = 98\%.$$

5) **b**; No. of students who scored more than 60% = 64.

$\therefore$ Percentage of students who scored more than 60% =

$$\frac{64}{108} \times 100 = 59.26\% = 59.3\% \text{ (approximately).}$$

Sample Questions

1) The following table shows the number of pages printed by 3 printers during 3 days.

Printers Days	X	Y	Z
Monday	130	140	210
Tuesday	110	145	160
Wednesday	180	90	218

What is the average number of pages printed by printer Z during the 3 days?

- a) 196 b) 194 c) 192 d) 190

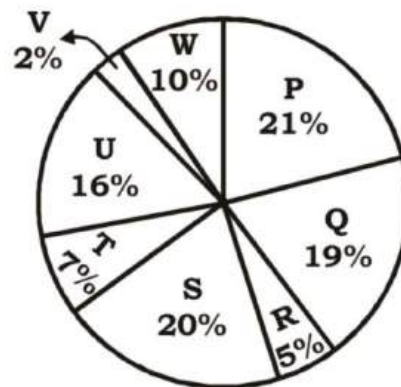
2) The table given below shows the number of spoon manufactured by five factories.

Factory	Spoon
P	100
Q	200
R	150
S	50
T	250

What are the ratio of number of spoon manufactured by P to the number of spoon manufactured by R?

- a) 3 : 2 b) 2 : 3 c) 14 : 1 d) 1 : 3

3) The pie chart given below shows the number of bike sold by 8 different companies. The total number of bike sold by all these 8 companies are 2000. Number of bikes sold by a particular company is shown as a percent of total number of bike sold by all these 8 companies.



What is the difference between the average number of bikes sold by P, Q, R and S and the average number of bikes sold by T, U, V and W?

- a) 150 b) 200 c) 175 d) 125

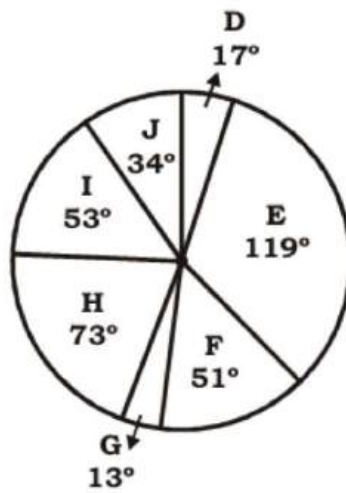
4) The table given below shows the number of bicycle sold by six companies.

Companies	Bicycle
A	60
B	90
C	120
D	165
E	180
F	120

What are the ratio of number of bicycle sold by B to the number of bicycle sold by D?

- a) 11 : 5 b) 11 : 6 c) 5 : 13 d) 6 : 11

5) The pie chart given below shows the sales of 7 different companies. The total sales of all these 7 companies are 2880. The sale of a particular company is shown in terms of degree with respect to the total sales of all these 7 companies.



The average sales of G and H is how much percent more than the average sales of D and I?

- a) 18.85% b) 24.25% c) 56.21% d) 22.86%

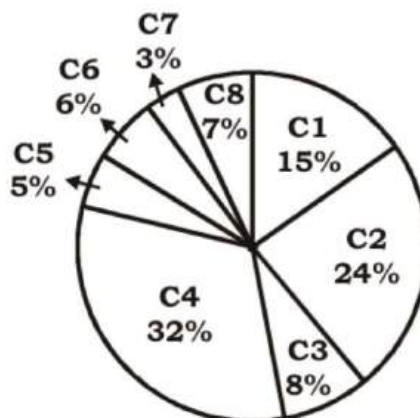
6) The table given below shows the production of beauty products by five different companies.

Companies	Production
P	80
Q	120
R	84
S	50
T	70

The production of beauty product by T is what percent of the production of beauty product by S?

- a) 40% b) 140% c) 240% d) 120%

7) The pie chart given below shows the population of 8 cities in a year. Total population of all the 8 cities is 30,000. Population in a particular city is shown as a percent of total population in all these 8 cities.



What is the average population (approximately) of cities C3, C5 and C7?

- a) 1200 b) 1800 c) 1600 d) 2000

8) The table given below shows the number of persons participating in a survey from 6 different states.

States	Persons
S1	100
S2	200
S3	400
S4	500
S5	600
S6	800

What is the ratio of number of person participating in a survey from state S3 to the number of person participating in a survey from state S4?

- a) 3 : 5 b) 5 : 4 c) 4 : 3 d) 4 : 5

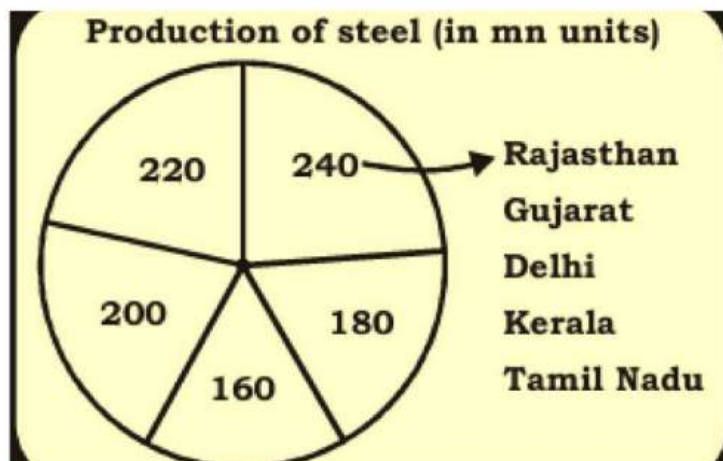
9) Study of the following table and answer the question below.

School Name	Total number of students enrolled	Percentage of enrolled students, opted Biology	Ratio of male to female students who opted Biology
A	900	30	7 : 8
B	400	36	5 : 7
C	1000	24	5 : 19
D	800	18	3 : 9

Find the ratio of the number of female students who opted biology in school B and school D.

- a) 7 : 9 b) 9 : 7 c) 3 : 5 d) 5 : 3

10) Study the given chart and answer the following question.



What is the central angle corresponding to the production steel by Kerala?

- a) 82° b) 65.8° c) 75.2° d) 72°

11) The table given below shows the income of two companies in 6 years

Years	Company 1	Company 2
P	750	850
Q	200	250
R	330	350
S	550	650
T	530	270
U	370	390

Which of the following statement is NOT correct?

I) The income of Company 1 in year P is 33.33 percent of the income of Company 2 in year Q.

II) The average income of Company 1 and Company 2 in year T is 400.

- a) Only 1 b) Both I and II
c) Neither I nor II d) Only II

12) The table given below shows the number of pencils sold by 7 shops.

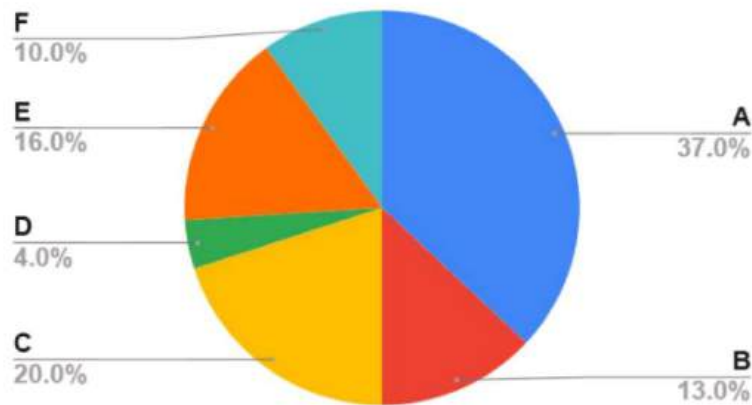
Shops	Pencils
S1	54
S2	68
S3	70
S4	120
S5	84
S6	130
S7	36

What is the ratio of number of pencils sold by S3 to the number of pencils sold by S6?

- a) 8 : 15 b) 13 : 7 c) 15 : 7 d) 7 : 13

13) The pie chart given below shows the production of 6 different factories. The total production of all these 6 factories are 22000.

The production of a particular factory is shown as a percent of total production of all these 6 factories.



P₁ = The value of average production of A, C, D and F.

P₂ = The difference between the production of B and E.

What is the value of (P₁ + P₂)?

- a) 2525 b) 6456 c) 3525 d) 4565

14) The following table gives the subscription of different schemes of a Mutual Fund Company over the months.

Months	V	W	X	Y	Z	Total
September	200	70	30	290	10	600
October	120	130	70	150	290	760
November	45	35	25	125	160	390
December	160	110	40	115	130	555
January	80	90	70	100	140	480
February	130	150	30	40	390	740

What is the difference in the subscription of scheme X between December and February?

- a) 10 crores b) 18 crores c) 8 crores d) 25 crores

15) The following table shows the marks (in percentages) obtained by six students in four different subjects in an examination.

Maximum marks in each subject are 100.

Student	SST	PHYSICS	CHEMISTRY	MATHS
A	92	90	90	80
B	90	80	85	85
C	80	80	65	70
D	85	80	82	75
E	80	75	75	85
F	90	90	90	85

What is the number of students whose overall percentage obtained is 80% and above?

a) 4

b) 5

c) 2

d) 3

Solutions

1) A

2) B

3) A

4) D

5) D

6) B

7) C

8) D

9) A

10) D

11) D

12) D

13) D

14) A

15) A

Algebra

$$1 + 2 + 3 + 4 + 5 + \dots + n = \frac{n(n+1)}{2}$$

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)(2n+1)}{6} \right)^2$$

$$\text{Sum of first } n \text{ odd numbers} = n^2$$

$$\text{Sum of first } n \text{ even numbers} = n(n+1)$$

$$(a+b)(a-b) = (a^2 - b^2)$$

$$(a+b)^2 = (a^2 + b^2 + 2ab)$$

$$(a-b)^2 = (a^2 + b^2 - 2ab)$$

$$(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$(a^3 + b^3) = (a+b)(a^2 - ab + b^2)$$

$$(a^3 - b^3) = (a-b)(a^2 + ab + b^2)$$

$$(a^3 + b^3 + c^3 - 3abc) = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac)$$

$$\text{When } a + b + c = 0, \text{ then } a^3 + b^3 + c^3 = 3abc$$

Quick Formulas

To find out son's age, use this formula –

If t_1 years earlier the father's age was x times that of his son. At present the father's age is y times that of his son. Then the present age of son will be?

$$\text{Son's Age} = t_1 \frac{(x-1)}{(x-y)}$$

If present age of the father is y times the age of his son. After t_2

years the father's age become z times the age of his son. Then the present age of son will be?

$$\text{Son's Age} = \frac{(z - 1)t_1}{(y - z)}$$

t_1 years earlier, the age of the father was x times the age of his son. After t_2 years, the father's age becomes x times the age of his son. Then the present age of son will be?

$$\text{Son's Age} = \frac{[(z - 1)t_2 + (x - 1)t_1]}{(x - z)}$$

$$\text{Son's or Daughter's Age} = \frac{[\text{Total ages} + \text{No. of years ago (Times - 1)}]}{(\text{Times} + 1)}$$

$$\text{Son's or Daughter's Age} = \frac{[\text{Total ages} - \text{No. of years ago (Times - 1)}]}{(\text{Times} + 1)}$$

Father : Son

Present Age = $x : y$

T years before = $a : b$

$$\text{Then, Son's age} = \frac{y \times T(a - b)}{\text{Difference of cross product}}$$

$$\text{And Father's age} = \frac{x \times T(a - b)}{\text{Difference of cross product}}$$

Sample Questions

1) The average age of a family of five members is 24. If the present age of the youngest member is 8 yr, then what was the average age of the family at the time of the birth of the youngest member?

- a) 20 b) 16 c) 12 d) 18

2) The sum of five numbers is 924. The average of first two numbers is 201.5 and the average of last two numbers is 196. What is the third number

- a) 133 b) 129 c) 122 d) 125

3) The average marks of 65 students in a class was calculated as

150. It was later realised that the marks of one of the students was calculated as 142, whereas his actual marks were 152. What is the actual average marks of the group of 65 students? (Rounded off to two digits after decimal)

- a) 151.25 b) 150.15 c) 151.10 d) 150.19

4) If the actual marks of two students were misread as 48 and 65 of the actual marks 72 and 61 respectively, then what would be the correct average?

- a) 68.5 b) 69 c) 69.5 d) 70

5) The average weight of A, B and C is 84 kg. If D joins the group, the average weight of the group becomes 80 kg. If another man E who weighs 3 kg more than D replaces A, then the average of B, C, D and E becomes 79 kg. What is the weight of A?

- a) 64 kg b) 72 kg c) 75 kg d) 80 kg

6) The average of 11 results is 50. If the average of first 6 results is 49 and that of last 6 is 52, find the 6th result

- a) 50 b) 52 c) 56 d) 60

7) The average of 30 observations is 45. If three new observations 42, 44 and 48 be added, find the new average.

- a) 42.9 b) 40.1 c) 42.4 d) 44.9

8) Average of two numbers is 14.5 and square root of their product is 10. What are the numbers?

- a) 25, 4 b) 20, 5 c) 10, 15 d) 10, 25

9) If average of 25 numbers is 30. If each no. decrease by 10. Then find new average of these no.

- a) 15 b) 20 c) 30 d) 40

10) A person divides his total route of journey into three equal parts and decides to travel the three parts with speeds of 40, 30 and 15 km/hr respectively. Find his average speed during the whole journey.

- a) 14 km/h b) 24 km/h c) 34 km/h d) 44 km/h

11) The average age of a lady and her daughter is 28.5. The ratio of their ages is 14: 5 respectively. What is the daughter's age?

- a) 12 years b) 15 years c) 18 years d) 16 years

12) The age of a man is 4 times that of his son. 5 yrs ago, the man was nine times as old as his son was at that time. What is the present age of the man?

- a) 28 years b) 32 years c) 40 years d) 42 years

13) After 5 yrs, the age of a father will be thrice the age of his son, whereas five years ago, he was 7 times as old as his son was. What are their present ages?

- a) 30 years b) 40 years c) 50 years d) 60 years

14) The ratio of the father's age to the son's age is 4: 1. The product of their ages is 196. What will be the ratio of their ages after 5 years?

- a) 7 : 3 b) 14 : 9 c) 11 : 4 d) 17 : 3

15) A man's age is 125% of what it was 10 years ago, but $83\frac{1}{3}\%$ of what it will be after 10 years. What is his present age?

- a) 30 years b) 40 years c) 50 years d) 60 years

Solutions

- | | | | |
|-------|-------|-------|-------|
| 1) B | 2) B | 3) B | 4) B |
| 5) C | 6) C | 7) D | 8) A |
| 9) B | 10) B | 11) B | 12) B |
| 13) B | 14) C | 15) C | |

Placement Paper 1

1. Ravi's salary was reduced by 25%. Percentage increase to be effected to bring the salary to the original level is

- a. 20%
- b. 25%
- c. $33\frac{1}{3}\%$
- d. 30%

Ans. C

2. A and B can finish a piece of work in 20 days .B and C in 30 days and C and A in 40 days. In how many days will A alone finish the job

- a. 48
- b. $34\frac{2}{7}$
- c. 44
- d. 45

Ans. A

3. A bag contains $x+5$ yellow balls, $2x+1$ blue balls and some red balls. If two balls are drawn one after another from the bag without replacement, then the probability of getting a red and blue ball is $\frac{1}{6}$. The total balls in the bag is $4(x+2)$. Find the number of red balls.

- a. 8
- b. 4
- c. 6
- d. 5

Ans. D

4. The taxi charges in a city consist of a fixed charge together with the fixed charge for the distance travelled in kilometers. When a person travels 72 km he pays 1107. He pays Rs. 898 for travelling 55 km. What will he have to pay for travelling 45 km?

- a. Rs. 826
- b. Rs. 740
- c. Rs. 693
- d. Rs. 738

Ans. D

5. A group of women can finish a piece of work in 50 days. In how

many days will $\frac{1}{3}$ rd of the number of women be able to finish $\frac{2}{3}$ rd of the work?

- a. 150 days
- b. 75 days
- c. 50 days
- d. 100 days

Ans. A

6. While presenting data in the form of positive numbers, I committed a mistake by mentioning the cube of the number. If the error due to this mistake was 800%, what was the number?

- a. 2
- b. 3
- c. 4
- d. 5

Ans. B

7. On selling cheeses at rs.340 per kg , a dairy farmer incurred a loss of 5% .how many kg of cheese where sold if the total loss was Rs. 85?

- a. 4.5
- b. 4.75
- c. 8
- d. 6

Ans. B

8. In a birthday party, there are 20 identical slices of cake that have to be distributed among four children. If every child is to get at least 2 slices, then in how many ways can this distribution be done?

- a. 455
- b. 514
- c. 550
- d. 600

Ans. A

9. The price of Machine D equals the sum of the prices of machine A, B and C whose price are in the ratio 2:3:4 respectively. If weights of A, B, C and D varies as square of its individual price and difference of weight of D and A, B and C together is 9880 kg. Then what is the weight of D?

- a. 15390 kg

- b. 14790 kg
- c. 15800 kg
- d. none of these

Ans. A

10. If 4:6 is the ratio of number of girls and boys in a computer coaching class. If 50% of girls and 40% of boys are degree holders then the percentage of the candidates who are non-degree holders is:

- a. 12%
- b. 38%
- c. 56%
- d. 49%

Ans. C

11. A and B start from the same point and cover equal distances. A travels by car and covers the distance in 3 hours with a speed of 50 km/h. The B travels by bus which stops for 10 mins after covering 10 kms. In how much time will the bus reach the destination if the speed of the bus is 40% less than the speed of the car?

- a. 6 h 40 min
- b. 7 h
- c. 7 h 30 min
- d. 7 h 20 min

Ans. D

12. A sum when lent at the rate of 1.5% p.a. simple interest for x years amounted to 17,600. When the same sum was lent at the rate of 18% p.a. The Simple interest for $(x+2.5)$ years, it amounted for 24,320. The value of x and the sum, respectively are

- a. 2.5 and 12500
- b. 2.5 and 12800
- c. 2 and 12500
- d. 3.5 and 12800

Ans. A

13. The mean and the variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 successes is:

- a. $37/256$
- b. $219/256$
- c. $128/256$

d. 28/256

Ans. D

14. After spending $\frac{2}{3}$ of his salary on rent and food items and $\frac{1}{4}$ th of the remaining on transportation, Rajesh has 6000 with him. How much did he pay on transportation?

a. 2000

b. 4000

c. 6000

d. 1000

Ans. D

15. In base 10 representation for a rupee 100 paise, then base 8 representation what is rupee value

a. 142

b. 143

c. 144

d. 145

Ans. C

16. Mr. Shah decided to walk down the escalator of a tube station. He found that if he walks down 26 steps, he requires 30 seconds to reach the bottom. However, if he steps down 34 stairs he would only require 18 seconds to get to the bottom. If the time is measured from the moment the top step begins to descend to the time he steps off the last step at the bottom, find out the height of the stair way in steps?

a. 46 steps

b. 47 steps

c. 48 steps

d. 49 steps

Ans. A

17. The average age of 10 members of a committee is the same as it was 4 years ago, because an old member has been replaced by a young member. Find how much younger is the new member ?

a. 39 years.

b. 40 years.

c. 41 years.

d. 42 years

Ans. B

18. The square of a two digit number is divided by half the number. After 36 is added to the quotient, this sum is then divided by 2. The digits of the resulting number are the same as those in the original number, but they are in reverse order. The ten's place of the original number is equal to twice the difference between its digits. What is the number?

- a. 43
- b. 44
- c. 45
- d. 46

Ans. C

19. A person travels 6km towards west, then travels 5 km towards north ,then finally travels 6km towards west. Where is he with respect to his starting position?

- a. 13 km east
- b. 13 km northeast
- c. 13 km northwest
- d. 13 km west

Ans. C

20. If A speaks the truth 80% of the times, B speaks the truth 60% of the times. What is the probability that they tell the truth at the same time

- a. 0.8
- b. 0.48
- c. 0.6
- d. 0.14

Ans. B

21. A monkey starts climbing up a tree 20ft. tall. Each hour, it hops 3ft. and slips back 2ft. How much time would it take the monkey to reach the top?

- a. 17 hours
- b. 18 hours
- c. 19 hours
- d. 20 hours

Ans. B

22. A certain type of mixture is prepared by mixing brand A at Rs.9 a kg. with brand B at Rs.4 a kg. If the mixture is worth Rs.7 a kg., how many kgs. brand A are needed to make 40kgs. of the mixture?

- a. 24kgs.
 - b. 25kgs.
 - c. 26kgs.
 - d. 27kgs
- Ans. A

23. Raj sold his bat at $x\%$ profit after giving a discount of $x\%$. The marked price was Rs. 2,400 more than the cost price. And the selling price was Rs. 900 more than the cost price. Find the value of $4x$.

- a. 100
- b. 50
- c. 40
- d. 200

Ans. A

24. The total number of students studying Science in institutes A and D is approximately what percentage more than the total number of male students studying in commerce in institutes B,C and E ? (correct to one decimal point) Discipline Ratio of male to female students Management 5:3 Engineering 2:3 Science 7:11 Arts 3:2 Computer Science 10:11 Commerce 8:7

- a. 42.8%
- b. 54.7%
- c. 35.4%
- d. 48.6%

Ans. B

25. A man ate 100 bananas in five days, each day eating 6 more than the previous day. How many bananas did he eat on the first day?

- a. 7
- b. 8
- c. 9
- d. 10

Ans. B

26. A group of women can finish a piece of work in 50 days.in how mny days will 1-3rd the number of women be able to finish 2-3rd of the work?

- a. 150 days
- b. 75 days

- c. 50 days
 - d. 100 days
- Ans. B

27. A sum invested at simple interest increased to rs 18375 and rs.20125 in 5 & 7 years, respectively .what was the interest (in rs) earned in 1 year?

- a. 976
- b. 875
- c. 925
- d. 950

Ans. B

28. I was supposed to completed a task in 12 h, but i took more time and the error was 6.25 percentage. how much time did i take to complete the task?

- a. 12 h 45 min
- b. 12h 20 min
- c. 12 h 15 min
- d. 12 h 30 min

Ans. A

29. age of 5 persons in a group is 18 years. the group as an additional numbers ,such that the new average is 25 years less than the age of the new members, what is the age (in years)of the new member?

- a.42
- b.38
- c.50
- d.48

Ans. C

30. X had taken a loan from y on simple interest at the rate of 5% per annum and x was supposed to pay back rs,23000 at the end of 3 years . he paid back rs.9450 and rs 4400 respectively ,at the end of the first and second year. How much (in rs) did x owe y at the end of third year?

- a.9150
- b.8050
- c.10665
- d.7500

Ans. A

Placement Paper 2

1. A car dealer sells varun a car at a 20% profit and then varun sells it to Deepak at a 15% profit, after using a car for 2 years Deepak sells it back to varun at a loss of 10%, whereas varun sells it back to the dealer at a loss of 5% the profit percentage that the dealer makes in this transaction is:

- a. 18.75%
- b. 16.82%
- c. 17.99%
- d. 15.62%

Ans. C

2. P is 5 times as efficient as Q in respect of doing a work .together they complete it in 20 days .what is the time of P?

- a. 26
- b. 30
- c. 28
- d. 24

Ans. D

3. They were 36 students in a class and their average height was 4 feet 6 inches.at the end of the year 4 students with an average height students got admitted to this class, their average height being 4 feet 6 inches, the average height in feet of students currently in

- a. $4 \frac{1}{6}$
- b. $4 \frac{3}{4}$
- c. $4\frac{1}{3}$
- d. $4\frac{2}{3}$

Ans. B

4. While presenting a data in the form of positive number i committed a mistake on monitoring the cube of the number if the error due to this mistake was 800% what was the number?

- a. 4
- b. 5
- c. 3
- d. 2

Ans. D

5. Lohit invested sum of money for 290 years at $x\%$ rate of interest per annum at simple interest and obtained the sum after 20 years the value of x is?

- a. 6
- b. 5
- c. 8
- d. 101

Ans. B

6. A retailer gives a discount of 20% on an item and yet makes a profit as 10% what is the ratio of the price of retailer?

- a. 7 : 12
- b. 1 : 2
- c. 8 : 11
- d. 4 : 5

Ans. D

7. X goes to the office by driving at $\frac{5}{7}$ times his usual speed .if he normally takes half an hour to reach is office by now many times will he be late?

- a. 12
- b. 10.5
- c. 13.5
- d. 8

Ans. A

8. You have 2 black shocks 3 white shocks and 5 brown shocks in your suitcase.it is dark and you pick out 3 shocks from the suitcase with out looking at the colour what is the probability that you do NOT have a pair of same colour?

- a. $\frac{1}{4}$
- b. $\frac{3}{10}$
- c. $\frac{1}{10}$
- d. $\frac{1}{5}$

Ans. C

9. In a music talent show has 100 contestants and 10000 viewers .each viewer picks their top 5 performers any performer who is nominated by at least 20 % of viewers get a prize what is the maximum number of prizes that would be awarded?

- a. 50
- b. 10

- c. 5
 - d. 25
- Ans. D

10. In case of after transaction there was a loss of 18.75% what was the ratio of the cost price to the selling price?

- a. 4 : 3
- b. 8 : 5
- c. 16 : 13
- d. 32 : 25

Ans. C

11. A set of mugs consists of 5 plastic mugs, each of a different designs and 2 glass mugs each of a different design in how many ways can these 7 mugs be arranged in a row if the glass mugs are separated from each other?

- a. 5040
- b. 3600
- c. 1440
- d. 4320

Ans. B

13. An investment earns 4 paisa per rupee invested if at the end of the year, the interested earned by an investment is rs.100 then the investment is equal to

- a. Rs.2000
- b. Rs.2200
- c. Rs.1000
- d. Rs.2500

Ans. D

14. A restaurant serves small pizzas of 6-inch diameter and regular pizzas of 9-inch diameter. If a small pizza is priced at ₹200, then what is a fair price for a larger pizza of the same type?

- a. ₹450
- b. ₹300
- c. ₹350
- d. ₹400.

Ans. A

15. The average age of five persons in a group is 18 years. The group has an additional member, such that the new average is 25

years less than the age of the new member. What is the age(in years) of the new member?

- a. 48
- b. 38
- c. 50
- d. 42

Ans. A

16. The incomes of 2 persons p ,q of the ratio is 4:7 and their expenses are in the ratio of 1:2 if each of the saves rs.2000 then the income of p in rs is :

- a. 8000
- b. 3690
- c. 2000
- d. 6500

Ans. A

17. A value of air cooler was rs 41500 in the year 2017,and it is depreciating at the rate of 12% per annum the value of it after two years was (rounded of in rs)

- a. Rs.36,100
- b. Rs.21,380
- c. Rs.32,138
- d. Rs.35,286

Ans. C

18. While presenting a data in the form of a positive number, I committed a mistake by mentioning the cube of the number. If the error due to this mistake was 800%, what was the number?

- a. 2
- b. 4
- c. 5
- d. 3

Ans. D

19. A retailer gives a discount of 20% on an item, and yet makes a profit of 10%. What is the ratio of the cost price to the list price?

- a. 7 : 12
- b. 8 : 11
- c. 4 : 5
- d. 1 : 2

Ans. B

20. There were 36 students in a class and their average height was 4 feet 6 inches. At the end of the year, 4 students, with an average height of 3 feet 3.5 inches left the class. Moreover, 6 new students got admitted to this class, their average height being 4 feet 9 inches. The average height, in feet, of the students currently in the class is:

- a. $4\frac{1}{6}$
- b. $4\frac{2}{3}$
- c. $4\frac{3}{4}$
- d. $4\frac{1}{3}$

Ans. B

21. X goes to the office by driving at $\frac{5}{7}$ times his usual speed. If he normally takes half an hour to reach his office by how many minutes will he be late?

- a. 12
- b. 8
- c. 10.5
- d. 13.5

Ans. A

22. The effective speed of a boat downstream is 50 km/h, and the effective speed upstream is 25 km/h. What is the speed of the stream?

- a. 8 km/h
- b. 12.5 km/h
- c. 10 km/h
- d. 15 km/h

Ans. B

23. In case of a transaction, there was a loss of 18.75%. What was the ratio of the cost price to the selling price?

- a. 4 : 3
- b. 8 : 5
- c. 32 : 25
- d. 16 : 13

Ans. D

24. A sum of ₹10,00,000 was invested in a scheme where it got compounded annually and the sum increased to ₹11,06,704 in 2 years. What is the rate of interest?

- a. 5.2%

- b. 5.4%
- c. 4.8%
- d. 4.4%

Ans. A

25. Mita and Geeta jointly start a small business Mita invests ₹15,000, whereas Geeta invests ₹10,000. After 2 months, Mita withdraws ₹2,500 from her investment in the business. After 8 more months, Mita again withdraws ₹2,500. Geeta does not withdraw any money in 1 year. After 1 year, they make a profit of ₹8,100 from their business. If they want to share this profit in the ratio of their money in the business, then how much money, in ₹, should Mita get as her share of the profit?

- a. 4400
- b. 4200
- c. 4500
- d. 4300

Ans. C

26. I was supposed to complete a task in 12 h, but I took more time and the error was 6.25%. How much time did I take to complete the task?

- A. 12 h 15 min
- B. 12 h 30 min
- C. 12 h 20 min
- D. 12 h 45 min

Ans. D

27. Ramesh can complete a work in 20 days. Mohan is 66.67% as efficient as Ramesh. Mohan and Ramesh work together. Ramesh leaves after working for some days. The remaining work is done by Mohan in 10 days. After how many days did Ramesh leave the work?

- a) 10 days
- b) 6.5 days
- c) 8.5 days
- d) 8 days

Ans. D

28. A sum of Rs 7500 amounts to Rs 9075 at 10% p.a in a certain time, when the interest is compounding annually. What is the amount in (Rs) of the sum at the same sum at the same rate of the

earlier time.

- a. 9412
- b. 9438
- c. 9680
- d. 9580

Ans. B

29. A car covers a distance of 420 km at a uniform speed. If the speed of the car is reduced by 20 km/h, it takes 2 hours more to cover the same distance. Find the original speed of the car.

- a) 50 km/h
- b) 60 km/h
- c) 70 km/h
- d) 80 km/h

Ans. C

30. The sum of the ages of a father and his son is 60 years. If the father is twice as old as the son, what is the son's age?

- a) 15 years
- b) 20 years
- c) 25 years
- d) 30 years

Ans. B

Placement Paper 3

1. In case of after transaction there was a loss of 18.75% what was the ratio of the cost price to the selling price?

- a. 4 : 3
- b. 8 : 5
- c. 16 : 13
- d. 32 : 25

Ans. C

2. What is the value of $(15^2 - 14^2)$?

- a. 29
- b. 1
- c. 28
- d. 15

Ans. C

3. Two numbers are in the ratio 3:4. If their sum is 140, what is the larger number?

- a. 80
- b. 60
- c. 90
- d. 100

Ans. A

4. If the circumference of a circle is 88 cm, what is the diameter of the circle? (Take $\pi = 22/7$)

- a. 14 cm
- b. 28 cm
- c. 44 cm
- d. 42 cm

Ans. B

5. A man can complete a work in 6 hours, and another man can complete the same work in 8 hours. If they work together, how long will it take to complete the work?

- a. 3 hours
- b. 3.43 hours
- c. 3.6 hours
- d. 4 hours

Ans. B

6. A man borrows ₹8000 at 10% per annum compound interest. What will be the amount due after 2 years?

- a. ₹9600
- b. ₹9680
- c. ₹9680
- d. ₹9680

Ans. C

7. What is the value of $(16.5 \times 2.5) \div 2$?

- a. 20.25
- b. 19.5
- c. 21.25
- d. 24.75

Ans. A

8. A pipe can fill a tank in 30 minutes, and another pipe can empty it in 45 minutes. If both pipes are opened together, how long will it take to fill the tank?

- a. 90 min
- b. 120 min
- c. 135 min
- d. 60 min

Ans. C

9. The sum of the first 15 natural numbers is:

- a. 120
- b. 105
- c. 125
- d. 135

Ans. D

10. A car travels 50 km at a speed of 40 km/h and the next 50 km at a speed of 60 km/h. What is the average speed for the entire journey?

- a. 48 km/h
- b. 50 km/h
- c. 52 km/h
- d. 54 km/h

Ans. B

11. A person buys a cycle for ₹1200 and sells it for ₹1500. What is his profit percentage?

- a. 20%
 - b. 25%
 - c. 30%
 - d. 35%
- Ans. B

12. What is the HCF of 36 and 60?

- a. 6
- b. 12
- c. 18
- d. 24

Ans. B

13. A sum of ₹5000 is invested for 3 years at 10% per annum simple interest. What is the total amount received after 3 years?

- a. ₹6500
- b. ₹5600
- c. ₹6500
- d. ₹5800

Ans. A

14. If the circumference of a circle is 88 cm, what is its diameter?

- a. 28 cm
- b. 35 cm
- c. 22 cm
- d. 14 cm

Ans. C

15. A sum of money becomes ₹14400 in 3 years at a certain rate of simple interest. If the same sum becomes ₹16000 in 5 years at the same rate, what is the sum?

- a. ₹8000
- b. ₹9000
- c. ₹12000
- d. ₹10000

Ans. D

16. Two numbers are in the ratio 2:3. If their sum is 40, what is the larger number?

- a. 16
- b. 24

- c. 20
 - d. 30
- Ans. B

17. A man spends 80% of his income and saves ₹4000. What is his total income?

- a. ₹16000
- b. ₹20000
- c. ₹15000
- d. ₹18000

Ans. A

18. What is the sum of the first 20 even numbers?

- a. 220
- b. 420
- c. 240
- d. 400

Ans. D

19. A man can row 20 km downstream in 4 hours and the same distance upstream in 5 hours. What is the speed of the man in still water?

- a. 5 km/h
- b. 4.5 km/h
- c. 5.5 km/h
- d. 4 km/h

Ans. A

20. The area of a square is 144 sq. m. What is the side length of the square?

- a. 12 m
- b. 14 m
- c. 16 m
- d. 10 m

Ans. A

21. A number is increased by 20% and then decreased by 15%. What is the net change?

- a. 2% increase
- b. 3% increase
- c. 2% decrease
- d. 3% decrease

Ans. D

22. What is the least number which when divided by 6, 9, and 12 leaves the same remainder of 3 in each case?

- a. 33
- b. 51
- c. 75
- d. 63

Ans. B

23. A train crosses a platform of length 120 meters in 18 seconds, running at a speed of 60 km/h. What is the length of the train?

- a. 160 m
- b. 120 m
- c. 100 m
- d. 140 m

Ans. A

24. The sum of three consecutive numbers is 96. What is the largest number?

- a. 31
- b. 33
- c. 32
- d. 34

Ans. B

25. A person sold an article at a loss of 10%. If he had sold it for ₹50 more, he would have gained 5%. What was the cost price?

- a. ₹300
- b. ₹350
- c. ₹400
- d. ₹450

Ans. C

26. A shopkeeper allows a discount of 10% on the marked price of an article and still gains 20%. If the marked price is ₹1800, what is the cost price?

- a. ₹1200
- b. ₹1300

- c. ₹1350
 - d. ₹1400
- Ans. D

27. A car travels at 60 km/h for the first 2 hours and at 80 km/h for the next 3 hours. What is the total distance covered by the car?

- a. 360 km
- b. 400 km
- c. 420 km
- d. 440 km

Ans. C

28. The average age of 5 students is 20 years. If a new student is added, the average becomes 21 years. What is the age of the new student?

- a. 22 years
- b. 25 years
- c. 26 years
- d. 24 years

Ans. D

29. The perimeter of a rectangle is 60 cm. If the length is 20 cm, what is the breadth?

- a. 15 cm
- b. 10 cm
- c. 8 cm
- d. 12 cm

Ans. B

30. A man walks at a speed of 4 km/h. How long will it take him to cover a distance of 18 km?

- a. 3.5 hours
- b. 4.5 hours
- c. 5 hours
- d. 4 hours

Ans. C

Placement Paper 4

1. A trader sells an article at a loss of 10%. If the selling price was Rs. 540, what was the cost price?

- a) Rs. 600
- b) Rs. 500
- c) Rs. 550
- d) Rs. 590

Ans. A

2. A train travels 240 km in 4 hours. What is the speed of the train?

- a. 40 km/hr
- b. 50 km/hr
- c. 60 km/hr
- d. 80 km/hr

Ans. C

3. The ratio of the ages of two brothers is 5:6. If the elder one is 24 years old, what is the age of the younger brother?

- a. 20 years
- b. 22 years
- c. 18 years
- d. 25 years

Ans. B

4. The cost price of 20 articles is equal to the selling price of 15 articles. What is the profit percentage?

- a. 33.33%
- b. 25%
- c. 20%
- d. 40%

Ans. A

5. A man buys an article for ₹800 and sells it for ₹960. What is the profit percentage?

- a. 15%
- b. 18%
- c. 20%
- d. 12.5%

Ans. C

6. What is the smallest number which when divided by 12, 15, and 20 leaves a remainder of 5 in each case?

- a. 65
- b. 125
- c. 245
- d. 305

Ans. D

7. If a number is decreased by 20% and then increased by 25%, the final value is what percent of the original value?

- a. 100%
- b. 105%
- c. 104%
- d. 110%

Ans. B

8. A sum of ₹1600 amounts to ₹1764 in 2 years at simple interest. What is the rate of interest?

- a. 5%
- b. 6%
- c. 7%
- d. 8%

Ans. C

9. If 4 men can do a job in 12 days, how many days will it take for 6 men to complete the same job?

- a. 8 days
- b. 6 days
- c. 10 days
- d. 7 days

Ans. A

10. Two pipes can fill a tank in 12 minutes and 15 minutes respectively. If both pipes are opened together, how long will it take to fill the tank?

- a. 6 min
- b. 5 min
- c. 6.67 min
- d. 7.5 min

Ans. C

11. A man covers a certain distance at 90 km/h and returns at 60 km/h. What is his average speed for the entire journey?

- a. 75 km/h
- b. 72 km/h
- c. 78 km/h
- d. 70 km/h

Ans. B

12. A car covers a distance of 540 km in 9 hours. What is its speed?

- a. 60 km/h
- b. 65 km/h
- c. 70 km/h
- d. 75 km/h

Ans. D

13. A boat goes 12 km downstream in 2 hours, and the same distance upstream in 3 hours. What is the speed of the boat in still water?

- a. 7 km/h
- b. 6.5 km/h
- c. 8 km/h
- d. 6 km/h

Ans. C

14. The length of a rectangle is increased by 20% and its breadth is decreased by 10%. What is the percentage change in its area?

- a. 8% increase
- b. 10% increase
- c. 8% decrease
- d. No change

Ans. A

15. A student scores 60% marks in an exam and fails by 12 marks. If the passing mark is 40%, what is the total mark?

- a. 200
- b. 180
- c. 150
- d. 120

Ans. A

16. If the ratio of the speeds of two cars is 3:4 and the first car covers 600 km in 12 hours, how long will it take the second car to cover 800 km?

- a. 15 hours
- b. 10 hours
- c. 12 hours
- d. 16 hours

Ans. C

17. A man borrows ₹5000 at 5% per annum simple interest for 3 years. How much interest does he have to pay?

- a. ₹750
- b. ₹600
- c. ₹850
- d. ₹500

Ans. A

18. If 12 men can complete a work in 8 days, how long will it take for 16 men to complete the same work?

- a. 6 days
- b. 7 days
- c. 5 days
- d. 4 days

Ans. A

19. A person sells two articles at ₹120 each. On one, he gains 20%, and on the other, he loses 20%. What is his overall gain or loss percentage?

- a. 2% loss
- b. 4% loss
- c. 2% gain
- d. No profit, no loss

Ans. B

20. A man's salary was increased by 10% and then decreased by 10%. What is the net change in his salary?

- a. 1% increase
- b. No change
- c. 1% decrease
- d. 2% decrease

Ans. C

21. The sum of the first 10 odd numbers is:

- a. 100
- b. 81
- c. 121
- d. 99

Ans. C

22. Two numbers are in the ratio 3:5. If their sum is 64, what is the larger number?

- a. 32
- b. 40
- c. 24
- d. 36

Ans. B

23. If the circumference of a circle is 44 cm, what is its radius?

- a. 7 cm
- b. 14 cm
- c. 10 cm
- d. 11 cm

Ans. A

24. A man walks at 5 km/h for 2 hours and at 4 km/h for the next 3 hours. What is his average speed for the entire journey?

- a. 4.4 km/h
- b. 4.2 km/h
- c. 4.6 km/h
- d. 4.8 km/h

Ans. B

25. A fruit seller had some apples. He sells 40% of them and has 420 apples left. How many apples did he have originally?

- a. 700
- b. 750
- c. 680
- d. 720

Ans. A

26. The average of 7 numbers is 35. If one number is excluded, the average becomes 33. What is the excluded number?

- a. 50

- b. 40
 - c. 60
 - d. 45
- Ans. A

27. A train 120 meters long is running at a speed of 54 km/h. How long will it take to cross a platform of 180 meters?

- a. 15 sec
- b. 20 sec
- c. 18 sec
- d. 25 sec

Ans. B

28. If the area of a square is 144 sq. cm, what is the perimeter of the square?

- a. 36 cm
- b. 24 cm
- c. 48 cm
- d. 12 cm

Ans. C

29. The ratio of two numbers is 2:3, and their LCM is 60. What is the larger number?

- a. 12
- b. 20
- c. 15
- d. 18

Ans. C

30. A man takes 6 hours to row 20 km downstream and comes back in 10 hours. What is the speed of the man in still water?

- a. 3 km/h
- b. 2 km/h
- c. 2.5 km/h
- d. 3.5 km/h

Ans. D

Placement Paper 5

1. A man walks 30 km at 6 km/h and the next 40 km at 8 km/h. What is his average speed for the entire journey?

- a. 7 km/h
- b. 6.5 km/h
- c. 7.2 km/h
- d. 7.5 km/h

Ans. C

2. If a number is divided by 13, the remainder is 5. What will be the remainder if the square of the number is divided by 13?

- a. 7
- b. 8
- c. 6
- d. 9

Ans. D

3. The cost price of an article is ₹500, and the selling price is ₹600. What is the profit percentage?

- a. 10%
- b. 20%
- c. 25%
- d. 30%

Ans. C

4. A train 180 meters long is running at a speed of 72 km/h. How long will it take to cross a man standing on the platform?

- a. 8 sec
- b. 9 sec
- c. 10 sec
- d. 11 sec

Ans. C

5. If a sum of money doubles itself in 5 years at simple interest, what is the rate of interest?

- a. 15%
- b. 20%
- c. 10%
- d. 25%

Ans. B

6. The length of a rectangle is twice its breadth. If the area is 200 sq. m, what is the length of the rectangle?

- a. 10 m
- b. 15 m
- c. 20 m
- d. 25 m

Ans. C

7. The difference between simple interest and compound interest for a sum of ₹5000 at 10% per annum for 2 years is:

- a. ₹25
- b. ₹50
- c. ₹75
- d. ₹100

Ans. B

8. If 10 workers can complete a task in 8 days, how many workers are required to complete the same task in 5 days?

- a. 15
- b. 16
- c. 12
- d. 18

Ans. A

9. What is the LCM of 18, 24, and 36?

- a. 72
- b. 144
- c. 216
- d. 96

Ans. B

10. A car covers 150 km in 3 hours. How long will it take to cover 300 km at the same speed?

- a. 6 hours
- b. 4.5 hours
- c. 5 hours
- d. 6.5 hours

Ans. A

11. The average of five consecutive numbers is 25. What is the largest number?

- a. 26

- b. 27
 - c. 28
 - d. 29
- Ans. B

12. A person buys an article for ₹450 and sells it at a loss of 10%. What is the selling price?

- a. ₹405
- b. ₹410
- c. ₹400
- d. ₹420

Ans. A

13. If a person's salary increases by 25% and then decreases by 20%, what is the net percentage change?

- a. 2% increase
- b. 3% increase
- c. 4% increase
- d. 5% decrease

Ans. B

14. A train traveling at 90 km/h crosses a bridge of length 150 meters in 20 seconds. What is the length of the train?

- a. 150 m
- b. 180 m
- c. 200 m
- d. 120 m

Ans. C

15. A man buys 5 dozen eggs for ₹300 and sells them at ₹7 per egg. What is his profit or loss percentage?

- a. 15% profit
- b. 20% profit
- c. 25% profit
- d. 30% profit

Ans. C

16. The ratio of the present ages of A and B is 5:7. After 6 years, the ratio will become 3:4. What is A's present age?

- a. 30 years
- b. 25 years

- c. 20 years
 - d. 15 years
- Ans. C

17. A sum of money becomes ₹1440 in 2 years at a certain rate of simple interest. If the same sum becomes ₹1560 in 3 years, what is the rate of interest?

- a. 10%
- b. 12%
- c. 15%
- d. 8%

Ans. B

18. The perimeter of a square is 64 cm. What is the area of the square?

- a. 256 sq. cm
- b. 144 sq. cm
- c. 196 sq. cm
- d. 225 sq. cm

Ans. A

19. A man covers a distance of 120 km at a speed of 40 km/h and another 180 km at a speed of 60 km/h. What is his average speed?

- a. 50 km/h
- b. 52 km/h
- c. 54 km/h
- d. 56 km/h

Ans. A

20. The HCF of two numbers is 24, and their product is 5760. What is the LCM?

- a. 240
- b. 288
- c. 480
- d. 360

Ans. B

21. A sum of ₹7500 amounts to ₹9000 in 3 years at simple interest. What is the rate of interest?

- a. 5%
- b. 6%

- c. 8%
 - d. 7%
- Ans. C

22. A car covers a distance of 180 km at a speed of 45 km/h. How long will it take to cover this distance?

- a. 3 hours
- b. 4 hours
- c. 5 hours
- d. 6 hours

Ans. C

23. If a sum doubles in 5 years under simple interest, how long will it take to triple?

- a. 10 years
- b. 7.5 years
- c. 8 years
- d. 12.5 years

Ans. A

24. The average of four numbers is 45. If three numbers are 50, 40, and 55, what is the fourth number?

- a. 35
- b. 30
- c. 25
- d. 20

Ans. A

25. A mixture contains milk and water in the ratio 3:2. If 10 liters of water is added, the ratio becomes 3:4. What is the original quantity of milk in the mixture?

- a. 15 liters
- b. 18 liters
- c. 12 liters
- d. 20 liters

Ans. A

26. A shopkeeper marks an article at ₹1500 and allows a discount of 10%. What is the selling price?

- a. ₹1300
- b. ₹1350

c. ₹1450

d. ₹1200

Ans. B

27. A train 200 meters long passes a pole in 10 seconds. What is the speed of the train?

a. 60 km/h

b. 72 km/h

c. 80 km/h

d. 90 km/h

Ans. B

28. The area of a rectangle is 300 sq. m, and its length is 20 m. What is the breadth?

a. 10 m

b. 12 m

c. 15 m

d. 18 m

Ans. C

29. A person invests ₹6000 at 10% per annum compound interest. What will be the amount after 2 years?

a. ₹7260

b. ₹7200

c. ₹7300

d. ₹7440

Ans. D

30. The sum of two numbers is 40, and their difference is 8. What is the product of the two numbers?

a. 384

b. 336

c. 320

d. 400

Ans. A

Tips to use Job Sites

Popular Job Sites: Job sites like Naukri.com, Monster.com, and TimesJobs.com are widely used for job searching and applications. They also serve as databases for recruiters and HR professionals looking to find candidates. Here are the most important tips for using these sites effectively when applying for jobs.

- 1. Regularly Update Your Resume:** Update your resume on a daily basis, even with minor changes, to ensure your profile stays current. Updating your resume between 12 am to 5 am can increase the chances of receiving calls from recruiters.
- 2. Avoid Mentioning Expected Salary:** Avoid stating your expected salary in your profile. Quoting above or below market standards can lead to rejection. Salary can be discussed during the HR interview round if you're shortlisted.
- 3. Complete Your Profile Information:** The more details you add to your profile, the better. Recruiters search based on many parameters, so aim for your profile completeness bar to reach close to 100%.
- 4. Write a Clear Profile Summary:** Your profile summary should briefly describe your skills, competencies, and objectives. Be concise yet informative to attract attention.
- 5. Craft a Precise Resume Headline:** Use a headline that reflects your skills and experience. Avoid generic titles like "Krishna's Resume" or "Experienced in software industry." Good examples include "Sun Certified Java Programmer" or "Microsoft Certified Professional."
- 6. Use a Professional Email ID:** Make sure your email ID is professional. For example, "jamesgosling@example.com" or "james.gosling@example.com" are ideal. Avoid using nicknames, random numbers, or dates in your email ID to appear more professional.
- 7. Verify Your Contact Details:** Make sure your email and mobile numbers are verified. Verified contact details increase your

chances of getting calls from recruiters.

8. Consider Using Video Resumes: A video resume can be a unique way to showcase your personality and emphasize key skills. It complements your traditional resume and helps you stand out to employers.

9. Set Up Job Alerts: Most job sites allow you to sign up for email alerts about newly available jobs matching your criteria. Utilize this feature to stay updated on new job postings.

10. Customize Your Resume for Each Company: Tailor your resume to each company by reading the job description and required skills. Highlight relevant skills to show that you're a great fit for the position.

11. Align Your Online Profile and Resume: Ensure there are no discrepancies between your online profile and the resume you upload. Upload your resume in MS Word format, as it's widely accepted.

12. Avoid Factual Errors: Double-check details such as date of birth and contact information before posting to prevent factual errors.

13. List Relevant Key Skills: Include technical skills, tools, certifications, and programming languages you've used in projects or academics. Freshers should mention at least 2-3 projects or internships to show practical experience.

14. Set Profile Visibility: Adjust your profile visibility based on your job-seeking status. Set it to "Active" if you're actively looking for a job.

15. Apply Once Per Job Posting: Only apply once for each job listing. Use the option to hide jobs you've already applied for to avoid redundancy.

16. Broaden Your Job Search Keywords: Expand your search using relevant keywords, tools, designations, and technologies in your field. For example, keywords in IT may include programming languages, databases, testing, and specific roles like "junior developer" or "database administrator."

HR Interviews

Before the Interview

Research the Company: Gather information about the company, its industry, and the competition.

Prepare Your 30-Second Pitch: Have a brief, personal introduction ready.

List Questions for the Interviewer: Prepare thoughtful questions to ask during the interview.

Gather References: Obtain permission from references to use their names and prepare a reference list.

Practice Your Interview Answers: Conduct a mock interview to refine your responses.

Match Your Qualities to Job Requirements: Identify qualities that align with the job and prepare examples for each.

Common Interview Questions

Tell me about yourself.

Why should we hire you?

What do you know about this company?

What are your goals?

What are your strengths and weaknesses?

Why do you want to work here?

What has been your most significant achievement?

How would your last boss describe you?

What are your salary expectations?

Q. Where do you see yourself in five years?

A. I see myself as a more developed and skilled professional with in-depth knowledge of key technologies. I aim to grow along with the company, contributing to its success and consistently giving my best.

Q. What are your expectations from company XYZ?

A. I expect to grow both as an individual and as a professional, enhancing my skills and technical knowledge while contributing to the company's objectives and reaching new heights.

Q. Why do you want to join XYZ?

A. XYZ's rapid growth and diverse tech stack present ideal opportunities for me to develop my technical skills across various fields, allowing me to make a meaningful contribution.

Items to bring to the Interview

- Several copies of your resume on quality paper.
- A list of your references.
- A notebook and pen in a folder for taking notes.
- A portfolio showcasing your work.
- Clear directions to the interview location.
- The interviewer's contact information.

Upon Arrival

- Arrive 15 minutes early.
- Announce yourself professionally to the receptionist.
- Engage with people around you while waiting.
- Check your appearance in the restroom.
- Greet your interviewer with a firm handshake, smile, and make eye contact.

During the Interview

- Focus on your key points without sounding rehearsed.
- Show interest in the company and ask relevant questions.
- Listen actively and engage the interviewer.
- Thank the interviewer and inquire about the next steps at the conclusion.
- Request a business card for follow-up.

After the Interview

Send a thank-you letter within 24 hours, emphasizing your fit for the role. Follow up two weeks later to express continued interest and check on their decision.

If you don't get the job, consider asking for feedback.

If there's no decision yet, reaffirm your interest and suitability for the role.

Resume Preparation

Your resume is a critical document for prospective employers. It should be professional, concise, and accurate, highlighting your key qualifications and experience. Ensure that it reflects your career path, skills, and achievements effectively.

Key Elements of an Effective Resume

Professional and Honest Content: Present your experience and skills truthfully. Avoid exaggeration and ensure the resume is straightforward.

Concise Format: Keep it brief and impactful. A well-organized, to-the-point resume is better than a lengthy one.

Originality: Write your resume in your own words, though it's fine to use samples for inspiration.

Tips for Crafting a Strong Resume

Select a Readable Format and Font: Choose an organized layout and a clear font (e.g., Verdana, Times New Roman, Arial) in sizes 10-12.

Use Relevant Keywords: Include terms relevant to your field to enhance searchability and appeal to hiring managers.

Clear, Effective Titles: Use specific titles that reflect your responsibilities and accomplishments (e.g., "Developed Java API" instead of "Software Development").

Organize by Sections: Divide content into sections (e.g., Work Experience, Education, Skills, Personal Profile) for easy reading.

Prioritize Important Details: Start each section with the most important and recent experiences.

Contact Information: Clearly list your phone number, email, and address.

Use Bullet Points and Short Sentences: Keep it simple and avoid long explanations.

Positive Language: Focus on accomplishments and abilities, avoiding negative phrasing.

Professional Goals: Briefly state your career aspirations.

Include Testimonials: Mention achievements with testimonials if available.

Limit Personal Details: Only provide personal information (like age) if explicitly requested.

Contents of a Good Resume

Personal Details: The most important detail is your name, mentioned in bold. Your contact information including email IDs and telephone numbers should be clearly written.

Summary: Writing an objective is a thing of the past. Instead, summarize your experience and skill set in one or two effective sentences. This would immediately grab the hiring manager's attention. E.g. Five years of experience in software quality assurance.

Past Work Experience: Include the details of your previous employment and briefly mention the significant projects that you were involved in. Mention the organization name and also the duration of your work. The order should be starting from recent to past.

Group Discussion

A. Content: Content refers to your subject knowledge and general awareness. Building knowledge through current affairs is essential.

- Read newspapers and magazines regularly.
- Watch videos on important and trending topics.

B. Delivery: Delivery focuses on how effectively you communicate your thoughts. Even with strong knowledge, if you struggle to communicate clearly, your points may not be impactful.

- Practice through mock GDs to improve delivery and build confidence.
- Simulate an actual GD environment by gathering a group, selecting a topic, and having a structured discussion.
- Role-play feedback: Assign a friend to play the recruiter and provide feedback on communication, argument quality, and body language.

Mock GDs can be enhanced by practicing previously asked or commonly discussed topics. Additionally, practice speaking in front of a mirror, using common phrases that are effective in GDs.

Body Language

Your body language is key to conveying confidence and openness. A positive posture reflects your engagement and comfort.

Ideal Body Language Tips:

1. Smile: A warm, friendly smile shows approachability.

- Do's: Smile while listening to others' views.
- Don'ts: Avoid grinning during serious discussions.

2. Eye Contact: Eye contact helps you engage with the group, showing confidence and attentiveness.

- While Speaking: Make equal eye contact with all group members to avoid making others uncomfortable.
- While Listening: Focus on the speaker to show respect and interest.

Listening Cues

- 3. Listening Cues:** Nods or quick eyebrow flashes acknowledge others' points, making them feel recognized.
- 4. Hands:** Place your hands on your thighs or the table in a relaxed manner.
 - Use natural gestures for emphasis but avoid overusing them.
 - Avoid gripping objects or excessive hand movement, as this shows stress.
- 5. Legs:** Position legs at a right angle, feet facing the group. Avoid shaking legs, as it can indicate nervousness.
- 6. Back:** Sit upright but stay relaxed. Leaning slightly forward when others speak shows engagement.

Things to be keep in Mind

1. Take Notes

Writing down key points mentioned by others is highly beneficial. This helps you track the discussion flow and allows you to respond with relevant supporting or countering points.

2. Be an Active Listener

Listening well contributes to a coherent discussion that can lead to a meaningful conclusion. This shows you as a team player. Try to integrate valid points from others into your responses to demonstrate understanding and collaboration.

3. Show Courtesy and Politeness

Group discussions simulate team interactions in a workplace, so team spirit and professionalism are essential. Even if you disagree, avoid condescension. Use respectful language like:

- "I understand your point, but from another perspective..."
- "Let's consider it from a different angle."

Avoid phrases like "You are wrong" or "I completely disagree," as they come off as negative.

4. Avoid Over-Emphasizing Your Opinion

While it's good to assert your point, don't repeat it excessively or force it on others. Share your perspective clearly, then let others contribute. Engage in the discussion constructively, building on your points without dominating.

5. Support Your Arguments with Data

Whenever possible, back up your statements with data or facts. Credible information enhances your argument's weight. However, ensure accuracy, as incorrect facts can diminish your credibility.

6. Focus on One Subtopic at a Time

Address each subtopic individually to keep the discussion organized. Avoid dumping all your points at once, as this can cause confusion.

7. Stay Calm and Composed

Maintain a level-headed attitude throughout. Losing your temper or becoming overly emotional will not serve you well. If a point challenges your values, address it calmly and respectfully instead of reacting emotionally.

Tackling tricking situations

A. Handling a "Fish Market" Situation

A “fish market” scenario happens when the group experiences a clash of opinions and the discussion becomes chaotic. Here’s how to manage this situation effectively:

- **Take on the Mediator Role:** Calmly step in to give each speaker their turn. Recruiters appreciate candidates who can manage high-energy situations calmly. However, avoid excessive interruptions, as this may come across as too dominant.
- **Stay Calm and Patient:** Rather than shouting, remain silent when the discussion becomes too loud. Take notes on key points and re-enter the discussion when things settle down.
- **Identify Allies:** Find group members who share similar views and collaborate to present your stance more effectively.
- **Use Facts and Data:** Support your opinions with relevant facts, helping to re-center the discussion and potentially assert your position.

B. Redirecting When the Discussion Goes Off-Topic

It’s common for discussions to lose focus, either by exploring a sub-topic in excessive detail or jumping between points without reaching conclusions. Here’s how to steer the group back:

- **Revisit or Propose a Structure:** Politely intervene to remind the group of the agreed structure or propose a new

- one. Summarize key points to bring everyone back on track.
- Use Re-directive Phrases: Try using phrases such as:
 - “We have covered these points... but according to our structure, we should now discuss...”
 - “I feel we haven’t concluded some points of our structure... perhaps we could alter our outline by...”
 - “It seems we’re moving away from the main topic of... I think we should now focus on discussing...”

C. When You’re Unfamiliar with the Topic

Finding yourself in a discussion topic you know little about can be intimidating. Here’s how to remain calm and contribute:

- Listen Actively: Focus on listening cues and take notes. This demonstrates engagement and attentiveness, even if your contributions are minimal. You can also add brief insights based on others’ points.
- Seek Clarifications: Politely ask your peers for clarifications to gain a better understanding, or summarize the discussion up to that point. If the discussion becomes unstructured, offering to mediate can work to your advantage.

Word of advice

Focus on Quality Over Quantity: Aim to raise relevant points and speak meaningfully rather than dominating the conversation. Trying to seize every opportunity to speak without adding value can come across as arrogant to the recruiter. It’s a myth that the candidate who initiates the discussion is automatically selected. While starting the GD is encouraged, it’s your contributions’ quality—not the frequency—that truly matters.

Work Toward a Consensus, Not a Victory: Remember, a GD is a collaborative discussion, not a debate. In a debate, each side aims to “win,” while in a GD, the goal is to share various perspectives and ideally reach a consensus. Avoid getting defensive or overly attached to your viewpoint, as prolonged arguments often detract from your performance, showing a lack of flexibility and teamwork.

Prioritize Logic Over Popular Opinion: Your stance on a topic doesn’t influence your selection; rather, it’s the logic, reasoning, and support you provide for your views that counts.

Avoid the misconception that agreeing with the popular opinion will improve your chances, or that disagreeing simply to stand out will make a good impression. What recruiters value is a well-reasoned perspective, whether it aligns with or differs from the majority.